

Deuteron with D-waves in the nuclear model with explicit mesons

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Master's thesis

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Colophon

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Abstract

I investigate the deuteron in the nuclear model with explicit mesons. The mesons, in this case a pion, are treated on equal footing with the nucleons. I consider only the one pion approximation where the physical deuteron is a superposition of a neutron-proton state and a nucleon-nucleon-pion state. The nucleons are said to be dressed with a virtual pion.

The model parameters are chosen to produce the binding energy and charge radius of the deuteron. Initially, I do not allow the individual nucleons to become dressed when they move far apart. This simplifies the energy spectrum and adds an extra parameter, making it easier to describe the bound state. I then slowly lift this constraint, allowing the nucleons to be far apart in a negative energy configuration, where one of the nucleons gets dressed by the pion. This moves the bound state threshold from zero down to the energy of this state, such that the difference down to the ground state will be the binding energy of the deuteron. This model has only two parameters: A nucleon-pion range and a coupling strength between the states in the superposition. Allowing only S-waves, the parameters cannot be tuned to the accepted values of both the binding energy and the charge radius. However, with the addition of D-waves, exactly one choice of parameters can produce both of these, but at the cost of the quadrupole moment being about twice as large as the experimental value.

Resume

Jeg undersøger deuteronen i kernemodellen med explicitte mesoner. Mesoner, som i dette tilfælde er pioner, optræder på lige fod med nukleonerne. Jeg nøjes med at undersøge bidraget fra en enkelt pion, hvorfor den fysiske deuteron er en superposition af en neutron-proton tilstand og en nukelon-nukleon-pion tilstand. Nukleonerne siges at være dressed med en virtuel pion.

Modellens parameter vælges således at deuteorn får den rigtige bindingsenergi og ladningsradius. Til at starte med tillader jeg ikke, at de individuelle nukleoner kan blive dressed når de er langt fra hinanden. Dette gør energispektraet mere simpelt, samt at modellen får en ekstra parameter, der gør det lettere at beskrive den bundne tilstand. Lige så stille fjerner jeg begrænsningen, så nukleonerne kan eksistere hver for sig i en negativ-energitilstand. Her er en af nukleonerne dressed med pionen. Det betyder, at denne tilstand er tærsklen for den bundne tilstand, og deuterons bindingsenergi kan findes som forskellen mellem de to laveste tilstande. Med denne ændring har modellen kun to parameter: en nukleon-pion vekselvirkningsafstand og en vekselvirkningsstyrke mellem de to tilstande i deuteronsuperpositionen. Hvis jeg kun tillader S-bølger, kan parametrene ikke tunes til at give både deuterons bindingsenergi og ladningsradius. Når jeg inkludere D-bølger i beskrivelsen er dette dog muligt, men kun for et enkelt valg af parametre, og på bekostning af, at quadropolmomentet bliver ca. dobbelt så stort som den eksperimentelle værdi.

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Introduction

The deuteron is the simplest bound nucleus, consisting of just one proton and one neutron. It is only natural to begin the investigation of nucleon-nucleon interactions here. It is held together by the nuclear force, mediated by mesons. As pions are the lightest mesons, they are responsible for the longest range part of the nuclear force[1].

Descriptions of meson-exchange traditionally include phenomenological one-boson-exchange potentials with model parameters chosen to fit experimental data[2]. In the nuclear model with explicit mesons (MEM), the mesons are treated as their own particles, just like the nucleons. The nucleons interact by emitting and absorbing mesons, described by a phenomenological coupling operator. The physical deuteron is a superposition of many nucleon-nucleon-meson states, where the number of mesons in each state is different. They are virtual, and under a potential barrier proportional to their mass.

As proof of concept, the model has already been used to describe the deuteron with the exchange of the scalar-isoscalar σ -meson in [3]. In this contribution, I consider the exchange of pions in the approximation where only the zero- and one-pion states are included in the superposition.

The model benefits from having essentially just two parameters, a nucleon-pion range and a coupling strength between the two states. The states are built as a linear combination of explicitly correlated Gaussians. The Gaussians act as trial functions, and in conjunction with the variational principle, are chosen to minimize the energy of the system. The trial functions can be multiplied by solid spherical harmonics to constitute wavefunctions with appropriate orbital angular momentum. I use this to include D-waves in the description of the deuteron.

CHAPTER 2

Formalism

This chapter serves to outline the formalism around the variational principle and how we generate suitable trial functions for it, to solve the Schrödinger equation.

2.1 Notation

In describing the theory and the model we use conveniently compacted notation that is consistent between systems with different numbers of particles.

For an N-particle system, the position vectors $\vec{r}_i$ will be placed in N-sized vectors

$$\mathbf{r} = (\vec{r}_1 \quad \vec{r}_2 \quad \dots \quad \vec{r}_N)^T. \quad (2.1)$$

These vectors have most of the same properties as normal vectors, but not all. The 3-dimensional position vectors will be denoted by a vector arrow, and the new vectors will be written in bold if they contain 3D-vectors, and normally if they only contain numbers. For example, the N-sized vector w_i that picks out a single particle vector ($w_i^T \mathbf{r} = \vec{r}_i$) will not be written in bold font. The bold vectors have the following properties:

$$\begin{aligned} \mathbf{a}^T \mathbf{b} &= \sum_{i=1}^N \vec{a}_i \cdot \vec{b}_i, \\ \mathbf{a}^T A \mathbf{b} &= \sum_{i=1}^N \sum_{j=1}^N \vec{a}_i \cdot \vec{b}_j A_{ij}, \end{aligned} \quad (2.2)$$

and while $(w_i^T \mathbf{r})(w_j^T \mathbf{r}) = \mathbf{r}^T (w_i w_j^T) \mathbf{r}$, the same is not true if w contained vectors.

For consistency, this notation will be used even for 1 dimensional systems.

Further, spin and isospin states (e.g. $|\uparrow\rangle$, $|p\rangle$) will only be written in Bra-Ket notation when appearing in matrix elements.

2.2 Variational principle

The model relies on the variational principle, stating that for an arbitrary wave function $|\Psi\rangle$ in the state space of the Hamiltonian, its energy is an upper bound to the system's ground state[4],

$$E = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle} \geq E_{\text{ground state}}. \quad (2.3)$$

This result can be exploited numerically by cleverly constructing trial wave functions that approximate the ground state of the system. The trial wavefunctions can be chosen as linear combinations of basis functions ψ_i ,

$$|\Psi_0\rangle = \sum_{i=1}^n c_i |\psi_i(\alpha_i)\rangle, \quad (2.4)$$

where ψ_i depends on some parameters α_i . The c_i coefficients can be found by applying H to the function and multiplying by $\langle \Psi_0 |$,

$$\sum_{i,j=1}^n c_i^* c_j \langle \psi_i | H | \psi_j \rangle = E \sum_{i,j=1}^n c_i^* c_j \langle \psi_i | \psi_j \rangle. \quad (2.5)$$

Isolating the energy leads to the equation

$$E = \frac{c^\dagger \mathcal{H} c}{c^\dagger \mathcal{N} c}, \quad (2.6)$$

where $\mathcal{H}$ and $\mathcal{N}$ are the Hamiltonian and overlap matrices of size $n \times n$, and c is an $n \times 1$ vector of the c_i coefficients. The matrices have the elements

$$\mathcal{H}_{ij} = \langle \psi_i | H | \psi_j \rangle, \quad \mathcal{N}_{ij} = \langle \psi_i | \psi_j \rangle. \quad (2.7)$$

ψ_i and ψ_j are not necessarily orthogonal, so $\mathcal{N}$ can have elements off the diagonal that are non-zero.

Minimizing eq. (2.6) results in the generalized eigenvalue problem

$$\mathcal{H}c = E\mathcal{N}c \quad (2.8)$$

that can be solved to yield the c_i coefficients and the corresponding energy. In general, there will be n solutions, and if they are sorted by energy, the energy of the k' th solution will be an upper bound of the exact energy of the k' th excited state of the system[5].

2.3 Explicitly correlated Gaussians

A suitable choice of basis functions is correlated Gaussians, which are Gaussian functions depending on the square of the distance between two particles,

$$\psi = \exp \left(- \sum_{i < j=1}^N \left(\frac{\vec{r}_i - \vec{r}_j}{b_{ij}} \right)^2 \right). \quad (2.9)$$

N is the number of particles, $\vec{r}_i$ is the coordinate of the i 'th particle and b_{ij} are variational parameters used to generate different basis functions[3, 5]. ψ can be written in the conveniently compacted form,

$$\psi = \exp \left(- \mathbf{r}^T A \mathbf{r} \right) = \langle \mathbf{r} | A \rangle, \quad (2.10)$$

where $\mathbf{r}$ is the N -component vector of particle position vectors,

$$\mathbf{r} = \begin{pmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vdots \\ \vec{r}_N \end{pmatrix}. \quad (2.11)$$

A is a symmetric positive-definite matrix given as

$$A = \sum_{i < j=1}^N \frac{w_{ij} w_{ij}^T}{b_{ij}^2}, \quad (2.12)$$

where w_{ij} picks out the relative vector between particles i and j , from $\mathbf{r}$, ie.

$$w_{ij}^T \mathbf{r} = \vec{r}_i - \vec{r}_j, \quad (2.13)$$

A variation of the correlated Gaussian is the shifted correlated Gaussian,

$$\psi = \exp(-\mathbf{r}^T A \mathbf{r} + \mathbf{s}^T \mathbf{r}), \quad (2.14)$$

where $\mathbf{s}$ contains variational parameters in the form of an N -sized vector of vectors, such that $\mathbf{s}^T \mathbf{r} = \sum_{i=1}^N \vec{s}_i \cdot \vec{r}_i$. The shifted Gaussians will not be used as basis functions in this project. However, they are quite useful for calculating the matrix elements between Gaussians.

The full wave function will need to be multiplied by other wave functions, such as isospin, spin, and orbital angular momentum wave functions, depending on the model. For now, I will focus on the angular momentum wave functions, as these

will influence the matrix elements directly. The basis functions can be extended by multiplying by solid spin-spherical harmonics[5, 6],

$$\psi \rightarrow \exp(-\mathbf{r}^T \mathbf{A} \mathbf{r}) \mathcal{Y}_{JLS}^M(\mathbf{r}), \quad (2.15)$$

where $\mathcal{Y}_{JLS}^M$ are found using Clebsch-Gordan coefficients[7],

$$\mathcal{Y}_{JLS}^M(\mathbf{r}) = (u^T \mathbf{r})^L \sum_{m_L, m_S} \langle J, M | L, m_L; S, m_S \rangle |S, m_S\rangle Y_{LM}(u^T \mathbf{r}). \quad (2.16)$$

where u is an N-dimensional vector of numbers. These numbers are variational parameters. In a one-dimensional system, u contains just one parameter and can be excluded as it will be absorbed into the c_i -coefficients of the expansion (2.4).

To evaluate the matrix elements between basis functions with non-zero orbital angular momentum, it can be advantageous to rewrite the spherical harmonics Y_{LM} as prefactors in the form $(\mathbf{a}^T \mathbf{r})$.

Consider, for example, a one-dimensional system where $\mathbf{r} = (\vec{r})$. The spherical harmonics Y_{LM} are then linear combinations of r_+ , r_- and r_z where $r_{\pm} = r_x \pm ir_y$ [8]. For example,

$$r^2 Y_{22} \propto r_+ r_+. \quad (2.17)$$

This can be achieved by letting $\mathbf{a}^T = (\vec{e}_+)^T = ((1 \ i \ 0))$,

$$\psi \rightarrow (\mathbf{a}^T \mathbf{r})(\mathbf{a}^T \mathbf{r}) \exp(-\mathbf{r}^T \mathbf{A} \mathbf{r}). \quad (2.18)$$

Generally, one prefactor represents P-wave Gaussians,

$$(\mathbf{a}^T \mathbf{r}) \exp(-\mathbf{r}^T \mathbf{A} \mathbf{r}), \quad (2.19)$$

where $\mathbf{a}$ can contain arbitrary vectors, and D-wave Gaussians will have two prefactors,

$$(\mathbf{b}^T \mathbf{r})(\mathbf{a}^T \mathbf{r}) \exp(-\mathbf{r}^T \mathbf{A} \mathbf{r}), \quad (2.20)$$

and so on.

Using explicitly correlated Gaussians as basis functions has several advantages: They have analytical matrix elements, which will speed up computation times. They form a complete basis in the limit where the number of them becomes very large. This allows the energy to converge to the exact eigenvalue.

The Gaussians are however not very good at approximating the wavefunction at large distances. This means that quantities related to the asymptotic behavior of the wave function are hard to get right. Instead, correlated exponential basis functions are a better choice for these. However the exponential functions suffer from not having analytical matrix elements for systems containing more than three particles[4, 5].

2.4 Matrix elements

The commonly used matrix elements will be listed in the section. A detailed derivation of most of them can be found in appendix A

Overlaps

Let $|A\rangle$ represent the correlated Gaussian with parameter A . The overlap between two Gaussians is

$$\langle B|A\rangle = \left(\frac{\pi^{\text{size}(A)}}{\det(A+B)} \right)^{3/2} \equiv M_0. \quad (2.21)$$

The expectation value of $(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})$ in an S-state is

$$\langle B|(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\rangle = \frac{1}{2} \mathbf{b}^\top R \mathbf{a} M_0 \equiv M_1, \quad (2.22)$$

where $R = (A+B)^{-1}$. This formula can also be used as the overlap of two P-wave Gaussians, although $\mathbf{b}$ then needs to be complex conjugated.

For $(w^\top \mathbf{r})(w'^\top \mathbf{r})$, where w, w' do not contain vectors,

$$\langle B|(w^\top \mathbf{r})(w'^\top \mathbf{r})|A\rangle = \frac{3}{2} w^\top R w' M_0. \quad (2.23)$$

The expectation value of any odd number of prefactors is 0 for unshifted Gaussians. An S-wave and a P-wave can therefore not overlap.

The expectation value of four prefactors is

$$\begin{aligned} \langle B|(\mathbf{d}^\top \mathbf{r})(\mathbf{c}^\top \mathbf{r})(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\rangle = \\ \frac{M_0}{4} \left((\mathbf{b}^\top R \mathbf{a})(\mathbf{d}^\top R \mathbf{c}) + (\mathbf{b}^\top R \mathbf{c})(\mathbf{d}^\top R \mathbf{a}) + (\mathbf{b}^\top R \mathbf{d})(\mathbf{c}^\top R \mathbf{a}) \right) \equiv M_2, \end{aligned} \quad (2.24)$$

while the expectation value of six prefactors is

$$\begin{aligned} \langle B|(\mathbf{f}^\top \mathbf{r})(\mathbf{e}^\top \mathbf{r})(\mathbf{d}^\top \mathbf{r})(\mathbf{c}^\top \mathbf{r})(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\rangle = \\ (M_0/8) [(\mathbf{a}^\top R \mathbf{b})(\mathbf{c}^\top R \mathbf{d})(\mathbf{e}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{b})(\mathbf{c}^\top R \mathbf{e})(\mathbf{d}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{b})(\mathbf{c}^\top R \mathbf{f})(\mathbf{d}^\top R \mathbf{e}) \\ + (\mathbf{a}^\top R \mathbf{c})(\mathbf{b}^\top R \mathbf{d})(\mathbf{e}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{c})(\mathbf{b}^\top R \mathbf{e})(\mathbf{d}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{c})(\mathbf{b}^\top R \mathbf{f})(\mathbf{d}^\top R \mathbf{e}) \\ + (\mathbf{a}^\top R \mathbf{d})(\mathbf{b}^\top R \mathbf{c})(\mathbf{e}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{d})(\mathbf{b}^\top R \mathbf{e})(\mathbf{c}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{d})(\mathbf{b}^\top R \mathbf{f})(\mathbf{c}^\top R \mathbf{e}) \\ + (\mathbf{a}^\top R \mathbf{e})(\mathbf{b}^\top R \mathbf{c})(\mathbf{d}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{e})(\mathbf{b}^\top R \mathbf{d})(\mathbf{c}^\top R \mathbf{f}) + (\mathbf{a}^\top R \mathbf{e})(\mathbf{b}^\top R \mathbf{f})(\mathbf{c}^\top R \mathbf{d}) \\ + (\mathbf{a}^\top R \mathbf{f})(\mathbf{b}^\top R \mathbf{c})(\mathbf{e}^\top R \mathbf{d}) + (\mathbf{a}^\top R \mathbf{f})(\mathbf{b}^\top R \mathbf{d})(\mathbf{c}^\top R \mathbf{e}) + (\mathbf{a}^\top R \mathbf{f})(\mathbf{b}^\top R \mathbf{e})(\mathbf{c}^\top R \mathbf{d})] \\ \equiv M_3. \end{aligned} \quad (2.25)$$

Kinetic energy

It will later be shown that for an N -sized system with coordinates $\vec{r}_i$ contained in a vector $\mathbf{r}$, the kinetic energy operator takes the form

$$\hat{K} = -\frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top}, \quad (2.26)$$

where K is an $N \times N$ diagonal matrix. The derivatives follow the same convention as normal vector/matrix derivatives, and $\partial/\partial \mathbf{r}$ is therefore a row vector.

The expectation value of $\hat{K}$ between two Gaussians is

$$\langle B | \hat{K} | A \rangle = 6 \text{Tr}(BKAR)M_0 \equiv LM_0. \quad (2.27)$$

For P-wave Gaussians,

$$\begin{aligned} \langle B | (\mathbf{c}^\top \mathbf{r}) \hat{K} (\mathbf{a}^\top \mathbf{r}) | A \rangle &= \\ L \langle B | (\mathbf{c}^\top \mathbf{r}) (\mathbf{a}^\top \mathbf{r}) | A \rangle + c^\top ((1 - RB)K(1 - AR) + RAKBR) \mathbf{a} M_0 & \quad (2.28) \\ \equiv LM_1 + c^\top K' \mathbf{a} M_0 \end{aligned}$$

and for D-wave Gaussians,

$$\begin{aligned} \langle B | (\mathbf{d}^\top \mathbf{r}) (\mathbf{c}^\top \mathbf{r}) \hat{K} (\mathbf{b}^\top \mathbf{r}) (\mathbf{a}^\top \mathbf{r}) | A \rangle &= \\ LM_2 + \frac{M_0}{2} \left[(\mathbf{a}^\top R \mathbf{c}) (\mathbf{d}^\top K' \mathbf{b}) + (\mathbf{a}^\top R \mathbf{d}) (\mathbf{c}^\top K' \mathbf{b}) \right. & \quad (2.29) \\ + (\mathbf{b}^\top R \mathbf{c}) (\mathbf{d}^\top K' \mathbf{a}) + (\mathbf{b}^\top R \mathbf{d}) (\mathbf{c}^\top K' \mathbf{a}) & \\ - (\mathbf{a}^\top R \mathbf{b}) (\mathbf{d}^\top [(1 - RB)KAR + RAK(1 - BR)] \mathbf{c}) & \\ \left. - (\mathbf{c}^\top R \mathbf{d}) (\mathbf{b}^\top [(1 - RA)KBR + RBK(1 - AR)] \mathbf{a}) \right] & \end{aligned}$$

Coulomb potential

For a $1/r$ potential, the expectation value in an S-state is

$$\langle B | \frac{1}{w^\top \mathbf{r}} | A \rangle = \frac{2}{\sqrt{\pi}} (w^\top R w)^{-1/2} M_0, \quad (2.30)$$

and in a P-state,

$$\langle B | (\mathbf{b}^\top \mathbf{r}) \frac{1}{w^\top \mathbf{r}} \mathbf{a}^\top \mathbf{r} | A \rangle = \frac{2}{\sqrt{\pi}} (w^\top R w)^{-1/2} M_1 - \frac{1}{3\sqrt{\pi}} (w^\top R w)^{-3/2} \mathbf{b}^\top R w w^\top R \mathbf{a} M_0. \quad (2.31)$$

2.5 Basis optimization

Choosing suitable parameters for the basis functions is essential for the energy to converge, whether this be to the ground state or one of the excited states. One method of choosing the parameters is by random sampling. One can employ pseudo or quasi-random sampling of the parameters while increasing the basis size until the energy converges. Using quasi-random sampling will provide a better quality basis, but using both to ensure they converge to the same value can be done to check for errors in the model. Depending on the system and the desired accuracy, this can require hundreds of randomly selected Gaussians before the energy is converged. It may be of advantage to only sample in a particular area where it is more likely to find Gaussians yielding a good improvement of the energy. This would typically depend on the size of the system.

Quasi-random sampling usually provides a better result because it intrinsically produces a low-discrepancy sequence of numbers, ie. there are no large unsampled areas, so long as the sequence is reasonably long[4, 9].

A problem with random sampling can occur if two Gaussians are nearly identical. Then the basis is not linearly independent. This can lead to the energy not being reliable or the eigenvalue problem not being solvable by conventional methods¹. There is also no benefit of having (nearly) identical elements, as they will not improve the energy. A quasi-random sequence is less likely to give overlapping Gaussians because of its low discrepancy.

To avoid this issue, when adding a new element, calculate its overlap with all the other basis elements (which will be done anyway when making $\mathcal{N}$), and if the overlap with one of the Gaussians is above a certain threshold of typically 95% to 99%, resample the element. The overlap between Gaussians is not normalized when using the matrix elements from the previous section. The new element is then kept if

$$\frac{\langle B|A \rangle}{\sqrt{\langle A|A \rangle \langle B|B \rangle}} \leq \text{threshold}. \quad (2.32)$$

In this project, I use both pseudo and quasi-random sampling to approximate the ground state of the system

As I use Python, solving the generalized eigenvalue is done using the `eigh` function from SciPy. Even though its time complexity is $O(N^3)$, it is not the most time consuming part of the stochastic sampling process, as SciPy code is compiled from C, C++, and Fortran[10, 11, 12, 13]. There is then no need to optimize solving the eigenvalue problem. Instead, it is evaluating the matrix elements that should be

¹ Methods that require $\mathcal{N}$ to be invertible. If two Gaussians are identical, $\mathcal{N}$ has identical rows and determinant 0.

optimized. It is beneficial to reuse calculated matrix elements whenever possible.

Another strategy for picking suitable basis elements is using minimization routines to find the Gaussians yielding the lowest energy. This can be done with a much smaller basis size than the random method. A problem with this kind of optimization is the presence of local minima. This can partially be circumvented by using a number of different starting positions.

Since many minimization routines will require the energy gradient to find the minimum, the Gaussian basis elements have another advantage, as the gradient is also analytic. However, in this project, I use the Nelder-Mead downhill simplex method, which does not rely on the gradient. This method benefits from being better at finding a minima if the initial guess is bad[4, 5, 14].

Using stochastic sampling is faster than minimizing the Gaussians, even though the number of Gaussians needed for minimization is several times less than for stochastic sampling. Minimization, however, benefits from being able to reuse optimized Gaussians when changing the model parameters. These Gaussians will likely converge to the ground state much quicker than randomly selected ones if the model parameters are only changed a bit.

Sometimes it may be necessary to find more than one energy level. This can prove a challenge, because Gaussians that describe one level well, might be ill suited to describe the others'. To circumvent this, Gaussians that give a good energy for one level can first be found. Then, finding Gaussians that fit the second level and adding them to the basis will result in a basis that gives a good approximation to the energy levels.

Another method for finding several energy levels is to minimize the sum of all those energies using the minimization routine.

Nuclear model with explicit mesons

The nuclear model with explicit mesons (MEM) is an interaction model where nucleons interact by emitting and absorbing mesons, and not by a phenomenological potential. The mesons are treated on equal footing with the nucleons. Unless enough energy is available, the mesons will be virtual under a potential barrier equal to their combined mass[3, 15].

In MEM, the physical deuteron is considered a superposition of a bare deuteron and two nucleons dressed with virtual pions,

$$\Psi = \begin{pmatrix} \psi_{\text{NN}} \\ \psi_{\text{NN}\pi} \\ \psi_{\text{NN}\pi\pi} \\ \vdots \end{pmatrix}. \quad (3.1)$$

The subscript indicates the number of pions in the given wave function. In the bare deuteron state (ψ_{NN}) the nucleons are a proton and a neutron.

The Hamiltonian describing the system is a matrix of the same dimension,

$$H = \begin{pmatrix} \hat{K}_{\text{N}} + \hat{K}_{\text{N}'} + m_{\text{N}} + m_{\text{N}'} & W^\dagger & 0 & \dots \\ W & H_{00} + \hat{K}_{\pi} + m_{\pi} & W^\dagger & \dots \\ 0 & W & H_{00} + \hat{K}_{\pi} + \hat{K}_{\pi'} + 2m_{\pi} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (3.2)$$

where $\hat{K}_{\text{N}}$, $\hat{K}_{\text{N}'}$ and $\hat{K}_{\pi}$ are the kinetic energy operators of the bare nucleons and the pion, m_{N} , $m_{\text{N}'}$ and m_{π} are the masses of the bare nucleons and the pion, and W and $W^\dagger$ are emission and absorption operators, that couples two states with a

difference of exactly one pion.

Since the energy produced by this Hamiltonian is $E = m_N + m_{N'} + E_B$, where E_B is the binding energy of the deuteron, the nucleon masses can be subtracted from the diagonal, and the resulting energy will then just be the binding energy.

The kinetic energy operator of the i 'th particle is

$$\hat{K}_i = -\frac{\hbar^2}{2m_i} \frac{\partial^2}{\partial \vec{r}_i^2}. \quad (3.3)$$

By assuming conservation of isospin, angular momentum, and parity in nuclear interactions, we construct the coupling operator to conserve these quantities. The simplest operator that does this is[1]

$$W = (\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1})F(r_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2})F(r_{\pi 2}). \quad (3.4)$$

$W^\dagger$ includes integrating over all space to get rid of the pion's coordinates[15]. The first term is responsible for emitting a pion from the first nucleon, and the second term for the second nucleon. $\vec{\tau}_i$ and $\vec{\sigma}_i$ are the vectors of Pauli matrices in the isospin space and spin space of the i 'th nucleon. $\vec{\pi}$ is the isovector of pions and $\vec{r}_{\pi i}$ is the relative vector between the pion and the i 'th nucleon. $F(r_{\pi i})$ is a short range form-factor that confines the pion to be close to a nucleon,

$$F(r_{\pi i}) = S_w f(r_{\pi i}) = S_w A e^{-r_{\pi i}^2/b_w^2}, \quad (3.5)$$

Where A is some normalization constant, chosen such that

$$\int_{\mathbb{R}^3} d^3 r_{\pi i} r_{\pi i}^2 f^2(r_{\pi i}) = 1, \quad (3.6)$$

resulting in

$$A = \left(4\pi \frac{3\sqrt{\pi} b_w^5}{2^{11/2}} \right)^{-1/2}. \quad (3.7)$$

This can be shown in several ways, eg. by using

$$r^2 e^{-2r^2/b^2} = \frac{b^3}{4} \frac{\partial}{\partial b} e^{-2r^2/b^2}, \quad (3.8)$$

and then carrying out the Gaussian integral over all spatial coordinates. This choice of normalization is twofold: It means S_w will be in units of energy, and it allows direct comparisons between different setups within MEM that also use this normalization, such as [16]. The $\vec{\tau} \vec{\pi}$ product is

$$\vec{\tau} \vec{\pi} = \tau_0 \pi^0 + \sqrt{2} \tau_+ \pi^- + \sqrt{2} \tau_- \pi^+, \quad (3.9)$$

where $\tau_{\pm}$ are raising and lowering operators for isospin states and τ_0 is the projection operator, ie. for the isospin states

$$p = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad n = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$\begin{aligned} \tau_+ p &= 0 & \tau_+ n &= p, \\ \tau_- p &= n & \tau_- n &= 0, \\ \tau_0 p &= p & \tau_0 n &= -n. \end{aligned}$$

W is the most general form of the pion creation operator, as it allows dressing a single nucleon in the state where the nucleons are far apart. To easier describe the bound state, W can be multiplied by $e^{-r_{\text{np}}^2/b_{\text{NN}}^2}$, where b_{NN} is another parameter of the model. This extra factor will confine the nucleons close to each other, so it cannot produce a state where they are far apart.

3.1 Coordinate transformations

If a system is not affected by external forces, and we only care about the intrinsic interaction, it can be advantageous to neglect the contributions of the center of mass motion. To do this, we make a coordinate transformation from the single particle coordinates $\mathbf{r} = (\vec{r}_1 \ \vec{r}_2 \ \dots \ \vec{r}_N)^T$ to a set of relative coordinates $\mathbf{x} = (\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_N)^T$, where $\vec{x}_N$ is the center of mass coordinate.

This means we can conveniently ignore the last element in $\mathbf{x}$ and thereby reduce the dimensionality of the system by 1, which will reduce the computational load[4]. One such set of relative coordinates is Jacobi coordinates. The first $N - 1$ transformations are given as

$$\vec{x}_i = \frac{1}{M_i} \sum_{j=1}^i m_j \vec{r}_j - \vec{r}_{i+1}, \quad (3.10)$$

where $M_i = \sum_{j=1}^i m_j$ is the sum of all the masses of the first i particles.

The transformations can be expressed using the matrix J , as

$$\mathbf{x} = J\mathbf{r}, \quad (3.11)$$

with

$$J = \begin{pmatrix} 1 & -1 & 0 & \dots & 0 \\ \frac{m_1}{M_2} & \frac{m_2}{M_2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{m_1}{M_{N-1}} & \frac{m_2}{M_{N-1}} & \frac{m_3}{M_{N-1}} & \dots & -1 \\ \frac{m_1}{M_N} & \frac{m_2}{M_N} & \frac{m_3}{M_N} & \dots & \frac{m_N}{M_N} \end{pmatrix}. \quad (3.12)$$

Under this transformation, the w vectors transform as

$$w \rightarrow U^\top w, \quad (3.13)$$

where $U = J^{-1}$ is the inverse of J . This transformation comes from the equation

$$w^\top \mathbf{r} = w^\top (U\mathbf{x}) = (U^\top w)^\top \mathbf{x}. \quad (3.14)$$

Removing the COM coordinate then corresponds to removing the last row of J and the last column of U .

The kinetic energy operators of eq. (3.3) also transform under the coordinate transformation, as they contain derivatives of the single particle coordinates. To see how they transform, consider that each diagonal element of the Hamiltonian in eq. (3.2) contains a sum of the kinetic energy operators of all the particles¹. The sum can be written in a more general way by defining the matrix K that has diagonal elements $(K)_{ii} = \frac{\hbar^2}{2m_i}$ and is zero otherwise. Then

$$\sum_i \hat{K}_i = -\frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top}, \quad (3.15)$$

where it becomes clear that the matrix K transforms as

$$K \rightarrow JKJ^\top, \quad (3.16)$$

when looking at the equation (with implied summation over all indices)[3]

$$\begin{aligned} \frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top} &= \frac{\partial}{\partial \vec{r}_i} K_{ij} \frac{\partial}{\partial \vec{r}_j} = \frac{\partial}{\partial \vec{x}_k} \frac{\partial \vec{x}_k}{\partial \vec{r}_i} K_{ij} \frac{\partial}{\partial \vec{x}_l} \frac{\partial \vec{x}_l}{\partial \vec{r}_j} \\ &= \frac{\partial}{\partial \vec{x}_k} J_{ki} K_{ij} J_{lj} \frac{\partial}{\partial \vec{x}_l} = \frac{\partial}{\partial \mathbf{x}} JKJ^\top \frac{\partial}{\partial \mathbf{x}^\top} \end{aligned} \quad (3.17)$$

¹ 'All the particles' refers to the particles in the specific state, ie. the first diagonal element of H only include a proton and a neutron

3.2 One-pion approximation

Including more pions leads to higher accuracy in calculations, but at a cost of significantly higher computational load, as the dimensionality of the system depends on the number of particles. Each pion also contributes increasingly less to the total wave function. I therefore start working in the one-pion approximation.

The physical deuteron's wave function is represented as the two-component wave function,

$$\Psi = \begin{pmatrix} \psi_{np} \\ \psi_{NN\pi} \end{pmatrix}. \quad (3.18)$$

These two wavefunctions will be referred to as the np-system and the π -system. Because the deuteron consists of two nucleons, which are fermions, its total wave function must be antisymmetric. Its wavefunction is mostly in an $S = 1, L = 0$ state, with only a few percent in an $S = 1, L = 2$ state[17]. I therefore initially will only consider the S-wave of the deuteron. The spin and angular momentum state of the S-wave is symmetric, so the isospin part of the deuteron wave function must be antisymmetric. The proton and neutron have isospin $I = 1/2$ and $I_3 = \pm 1/2$, so the deuteron is in an isospin singlet state. The wave function for the bare deuteron can therefore be chosen as

$$\psi_{np} = \frac{pn - np}{\sqrt{2}} \uparrow\uparrow \sum_{i=1}^{n_d} c_i e^{\mathbf{r}_d^T A_i^d \mathbf{r}_d}, \quad (3.19)$$

where c_i are the coefficients to be found by solving the generalized eigenvalue problem of eq. (2.8) and A_i^d are the correlation matrices of the correlated Gaussians for the np-state. n_d is the number of Gaussians in this state. The vector $\mathbf{r}_d$ is the transformed single particle vector,

$$\mathbf{r}_d = J\mathbf{r} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} \vec{r}_p \\ \vec{r}_n \end{pmatrix} = (\vec{r}_p - \vec{r}_n) = (\vec{r}_{np}). \quad (3.20)$$

Because the transformation removes one of the dimensions, the matrices A_i^d are just numbers but will be kept as 1×1 matrices for consistency.

The Schrödinger equation for the entire system reads

$$H\Psi = E\Psi \quad (3.21)$$

where E is just the binding energy because H has had the nucleon masses subtracted from its diagonal. The Schrödinger equation of this system gives two equations, the second being

$$\left((\vec{r}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1})F(r_{\pi 1}) + (\vec{r}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2})F(r_{\pi 2}) \right) \psi_{np} + \left(-\frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^T} + m_\pi \right) \psi_{NN\pi} = E \psi_{NN\pi}, \quad (3.22)$$

where now, $\mathbf{r} = (\vec{r}_N \ \vec{r}_{N'} \ \vec{r}_\pi)^\top$. The equation indicates that $\psi_{NN\pi}$ must have the same spin-isospin structure as $W\psi_{np}$. A possible choice could be

$$\psi_{NN\pi} = \left((\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \right) \frac{pn - np}{\sqrt{2}} \uparrow\uparrow \sum_{j=1}^{n_\pi} = c_j e^{\mathbf{r}_\pi^\top A_j^\pi \mathbf{r}_\pi}. \quad (3.23)$$

The vector $\mathbf{r}_\pi$ (not to be confused with the pion coordinate $\vec{r}_\pi$) is

$$\mathbf{r}_\pi = J\mathbf{r} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{m_N}{m_N + m_{N'}} & \frac{m_{N'}}{m_N + m_{N'}} & -1 \end{pmatrix} \begin{pmatrix} \vec{r}_N \\ \vec{r}_{N'} \\ \vec{r}_\pi \end{pmatrix} = \begin{pmatrix} \vec{r}_{N'N} \\ \vec{r}_{\pi N'N} \end{pmatrix}, \quad (3.24)$$

where $\vec{r}_{\pi N'N} = \frac{m_N \vec{r}_N + m_{N'} \vec{r}_{N'}}{m_N + m_{N'}} - \vec{r}_\pi$.

The kinetic energy operators for the two subsystems are

$$\hat{K}_d = -\frac{\partial}{\partial \mathbf{r}_d} K_d \frac{\partial}{\partial \mathbf{r}_d^\top}, \quad \hat{K}_\pi = -\frac{\partial}{\partial \mathbf{r}_\pi} K_\pi \frac{\partial}{\partial \mathbf{r}_\pi^\top}, \quad (3.25)$$

with

$$K_d = \left(\frac{\hbar^2}{2\mu_{np}} \right) \quad K_\pi = \begin{pmatrix} K_d & 0 \\ 0 & \frac{\hbar^2}{2\mu_{NN'\pi}} \end{pmatrix}, \quad (3.26)$$

where the reduced masses are

$$\mu_{np} = \frac{m_n m_p}{m_n + m_p} \quad \mu_{NN'\pi} = \frac{m_\pi (m_N + m_{N'})}{m_\pi + m_N + m_{N'}}. \quad (3.27)$$

The Hamiltonian then looks like

$$H = \begin{pmatrix} \hat{K}_d & W^\dagger \\ W & \hat{K}_\pi + m_\pi \end{pmatrix}. \quad (3.28)$$

Matrix elements in the one pion approximation

The Hamiltonian and overlap matrices for this system are given as the 2×2 block matrices,

$$\mathcal{H} = \begin{pmatrix} \langle \psi_i^d | \hat{K}_{np} | \psi_j^d \rangle & \langle \psi_i^d | W^\dagger | \psi_j^\pi \rangle \\ \langle \psi_i^\pi | W | \psi_j^d \rangle & \langle \psi_i^\pi | \hat{K}_\pi + m_\pi | \psi_j^\pi \rangle \end{pmatrix}, \quad (3.29)$$

$$\mathcal{N} = \begin{pmatrix} \langle \psi_i^d | \psi_j^d \rangle & 0 \\ 0 & \langle \psi_i^\pi | \psi_j^\pi \rangle \end{pmatrix}, \quad (3.30)$$

where

$$|\psi_i^d\rangle = \frac{pn - np}{\sqrt{2}} \uparrow\uparrow |A_i^d\rangle, \quad (3.31)$$

$$|\psi_i^\pi\rangle = \left((\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \right) \frac{pn - np}{\sqrt{2}} \uparrow\uparrow |A_i^d\rangle, \quad (3.32)$$

and where $\langle \psi_i^d | \hat{K}_{np} | \psi_j^d \rangle$ symbolises an $n_d \times n_d$ matrix with elements $\langle \psi_i^d | \hat{K}_{np} | \psi_j^d \rangle$. Likewise, the other inner products also symbolises matrices. In W, I will include the restriction on the inter-nucleon distance, which can always be removed by setting it large, or just omitting it.

Overlaps

The spin and isospin states are orthonormal, so

$$\langle \psi_i^d | \psi_j^d \rangle = \langle A_i^d | A_j^d \rangle. \quad (3.33)$$

In the $_{NN}\pi$ state, the τ and σ matrices affect the spin and isospin states,

$$\begin{aligned} \langle \psi_i^\pi | \psi_j^\pi \rangle &= \left\langle A_i^\pi \left| \uparrow\uparrow \right| \frac{\langle pn | - \langle np |}{\sqrt{2}} \left((\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \right)^\dagger \right. \\ &\quad \times \left. \left((\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \right) \frac{|pn\rangle - |np\rangle}{\sqrt{2}} \left| \uparrow\uparrow \right| A_j^\pi \right\rangle. \end{aligned} \quad (3.34)$$

Multiplying the two parenthesis gives 4 terms

$$\begin{aligned} &(\vec{\tau}_1 \vec{\pi})^\dagger (\vec{\sigma}_1 \vec{r}_{\pi 1})^\dagger (\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})^\dagger (\vec{\sigma}_2 \vec{r}_{\pi 2})^\dagger (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \\ &+ (\vec{\tau}_1 \vec{\pi})^\dagger (\vec{\sigma}_1 \vec{r}_{\pi 1})^\dagger (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) + (\vec{\tau}_2 \vec{\pi})^\dagger (\vec{\sigma}_2 \vec{r}_{\pi 2})^\dagger (\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}). \end{aligned} \quad (3.35)$$

For the first two terms, we use that²

$$(\vec{\tau} \vec{\pi})^\dagger (\vec{\tau} \vec{\pi}) = \pi_0^\dagger \pi_0 + \pi_+^\dagger \pi_+ + \pi_-^\dagger \pi_- = 3, \quad (3.36)$$

$$(\vec{\sigma} \vec{r})^\dagger (\vec{\sigma} \vec{r}) = r^2, \quad (3.37)$$

and for the last two terms,

$$\frac{1}{2} \left(\langle pn | - \langle np | \right) (\vec{\tau}_1 \vec{\pi})^\dagger (\vec{\tau}_2 \vec{\pi}) \left(|pn\rangle - |np\rangle \right) = -3, \quad (3.38)$$

$$\langle \uparrow\uparrow | (\vec{\sigma}_1 \vec{r}_{\pi 1})^\dagger (\vec{\sigma}_2 \vec{r}_{\pi 2}) | \uparrow\uparrow \rangle = r_{\pi 1}^z r_{\pi 2}^z, \quad (3.39)$$

where $r_{\pi 1}^z$ is the z -component of $\vec{r}_{\pi 1}$. We can exploit that the expectation value of the last two terms are the same to write the total overlap as

$$\langle \psi_i^\pi | \psi_j^\pi \rangle = 3 \langle A_i^\pi | r_{\pi 1}^2 + r_{\pi 2}^2 - 2r_{\pi 1}^z r_{\pi 2}^z | A_j^\pi \rangle. \quad (3.40)$$

² Equations (3.36) through (3.39) can be shown by explicitly applying the σ and τ matrices on the spin and isospin states. It can be advantageous to use $\vec{\sigma} \vec{r} = \sigma_0 r^z + \sigma_+ r^- + \sigma_- r^+$ where $r^\pm = r^x \pm ir^y$.

To use the matrix elements from Chapter 2, we need to apply the Jacobi transformation to the $\vec{r}_{\pi i}$ vectors. If $\mathbf{r} = (\vec{r}_N \ \vec{r}_{N'} \ \vec{r}_\pi)^\top$, then $\vec{r}_{\pi i} = w_{\pi i}^\top \mathbf{r} \rightarrow w_{\pi i}^\top \mathbf{r}_\pi$ where

$$w_{\pi 1} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \rightarrow U^\top w_{\pi 1} \quad w_{\pi 2} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \rightarrow U^\top w_{\pi 2}. \quad (3.41)$$

The z -component of the vectors are

$$r_{\pi 1}^z = \vec{e}_z \cdot \vec{r}_{\pi 1} = (\vec{e}_z w_{\pi 1})^\top \mathbf{r}_\pi = \mathbf{a}^\top \mathbf{r}_\pi \quad r_{\pi 2}^z = (\vec{e}_z w_{\pi 2})^\top \mathbf{r}_\pi = \mathbf{b}^\top \mathbf{r}_\pi. \quad (3.42)$$

So the overlap can be evaluated using the formula of the matrix elements as

$$\langle \psi_i^\pi | \psi_j^\pi \rangle = 3 \frac{1}{2} \langle A_i^\pi | A_j^\pi \rangle \left(3w_{\pi 2}^\top R w_{\pi 1} + 3w_{\pi 2}^\top R w_{\pi 2} - 2\mathbf{a}^\top R \mathbf{b} \right) \equiv M_\pi. \quad (3.43)$$

Hamiltonian

For the np-subsystem, the expectation value of the kinetic energy operator is

$$\langle \psi_i^d | \hat{K}_d | \psi_j^d \rangle = LM_0. \quad (3.44)$$

For the $\text{NN}\pi$ -subsystem, the matrix element will contain four terms as in eq. (3.35), but with $\hat{K}_\pi$ in all of them. The matrix element then reduces to

$$\langle \psi_i^\pi | \hat{K}_\pi | \psi_j^\pi \rangle = 3 \left\langle A_i^\pi \left| r_{\pi 1}^2 \hat{K}_\pi + r_{\pi 2}^2 \hat{K}_\pi - r_{\pi 1}^z \hat{K}_\pi r_{\pi 2}^z - r_{\pi 2}^z \hat{K}_\pi r_{\pi 1}^z \right| A_j^\pi \right\rangle. \quad (3.45)$$

Like before, using $\vec{r}_{\pi i} = w_{\pi i}^\top \mathbf{r}_\pi$ together with eq. (2.28), we find

$$\langle \psi_i^\pi | \hat{K}_\pi | \psi_j^\pi \rangle = LM_\pi + \left(9w_{\pi 2}^\top K' w_{\pi 1} + 9w_{\pi 2}^\top K' w_{\pi 2} - 3\mathbf{a}^\top K' \mathbf{b} - 3\mathbf{b}^\top K' \mathbf{a} \right). \quad (3.46)$$

The coupling elements will also resemble the overlap of the $\text{NN}\pi$ states,

$$\begin{aligned} \langle \psi_i^\pi | W | \psi_j^d \rangle &= 3 \langle A_i^\pi | r_{\pi 1}^2 F(r_{\pi 1}) + r_{\pi 2}^2 F(r_{\pi 2}) \\ &\quad - r_{\pi 1}^z r_{\pi 2}^z (F(r_{\pi 1}) + F(r_{\pi 2})) | A_j^d \rangle. \end{aligned} \quad (3.47)$$

For this we need to evaluate

$$\begin{aligned} \langle A_i^\pi | F(r_{\pi l}) | A_j^d \rangle &= S_w \langle A_i^\pi | e^{-r_{\pi l}^2/b_w^2} e^{-r_{\text{np}}^2/b_{\text{NN}}^2} | A_j^d \rangle \\ &= S_w \langle A_i^\pi | B_j^l \rangle, \end{aligned} \quad (3.48)$$

where

$$B_j^l = \begin{pmatrix} A_j^d + \frac{1}{b_{\text{np}}^2} & 0 \\ 0 & 0 \end{pmatrix} + \frac{w_{\pi l} w_{\pi l}^\top}{b_{\text{NN}}^2}, \quad (3.49)$$

and where the normalisation of W has been omitted due to it being an ugly factor. Then,

$$\begin{aligned} \langle \psi_i^\pi | W | \psi_j^d \rangle = 3S_w \frac{1}{2} & \left(3w_{\pi 2}^\top R^1 w_{\pi 1} \langle A_i^\pi | B_j^1 \rangle + 3w_{\pi 2}^\top R^2 w_{\pi 2} \langle A_i^\pi | B_j^2 \rangle \right. \\ & \left. - \mathbf{b}^\top R^1 \mathbf{a} \langle A_i^\pi | B_j^1 \rangle - \mathbf{a}^\top R^2 \mathbf{b} \langle A_i^\pi | B_j^2 \rangle \right), \end{aligned} \quad (3.50)$$

where $R^l = (A_i^\pi + B_j^l)^{-1}$.

Charge radius

One of the known quantities this model will try to reproduce is the charge radius of the deuteron, defined by[3]

$$R_c^2 = \sum_{k=1}^N Z_k \langle r_k^2 \rangle \quad (3.51)$$

where Z_k is the unit charge of particle k . Only the proton and some of the pions are charged, however the pions do not contribute to the charge radius, as the π^+ cancels out the π^- . This is shown in the results section. For the π -system, we can assume one of the nucleons have a positive charge. The system is in a superposition of a $np\pi^0$, $nn\pi^+$ and a $pp\pi^-$ state, so there is an average of one proton in each state, and I will show that their total contribution to the charge radius is the same as assuming one of the nucleons is charged.

The expectation value of the position vector of particle k is,

$$\begin{aligned} \langle r_k^2 \rangle = \langle \mathbf{r}^\top w_k w_k^\top \mathbf{r} \rangle &= \sum_{i,j=1}^{n_d} c_i c_j \langle A_i^d | \mathbf{r}_d^\top w_k w_k^\top \mathbf{r}_d | A_j^d \rangle \\ &+ \sum_{i,j=1}^{n_\pi} c_i c_j \langle \psi_i^\pi | \mathbf{r}_\pi^\top w_k w_k^\top \mathbf{r}_\pi | \psi_j^\pi \rangle. \end{aligned} \quad (3.52)$$

where w_k is the vector that picks out the position vector of particle k . It is transformed by $U_d^\top$ or $U_\pi^\top$, depending on what state it appears in. The inner product in the second term can be reduced to

$$\langle \psi_i^\pi | r_k^2 | \psi_j^\pi \rangle = 3 \langle A_i^\pi | r_{\pi 1}^2 r_k^2 + r_{\pi 2}^2 r_k^2 - 2r_{\pi 1}^z r_{\pi 2}^z r_k^2 | A_j^\pi \rangle, \quad (3.53)$$

which can be evaluated by the formula for overlap between to D-waves, by letting

$$\vec{r}_{\pi i} = w_{\pi i}^\top \mathbf{r}_\pi \quad \vec{r}_k = w_k^\top \mathbf{r}_\pi. \quad (3.54)$$

3.3 D-waves

The deuteron is in a $J = 1, M = 1$ angular momentum state and its wavefunction is a mixture of 3S_1 and 3D_1 waves. It is dominated by the S-wave, with only about 4-6% D-wave[1, 17]. To include the D-wave in this model of the deuteron, the ψ_{np} state is extended,

$$\begin{aligned} \psi_{np} &\rightarrow \psi_{np}^{(S\text{-wave})} + \psi_{np}^{(D\text{-wave})} \\ &= \frac{pn - np}{\sqrt{2}} \mathcal{Y}_{101}^1 \sum_{i=1}^{n_S} c_i e^{-\mathbf{r}_d^T A_i^d \mathbf{r}_d} + \frac{pn - np}{\sqrt{2}} \mathcal{Y}_{121}^1 \sum_{j=1}^{n_D} c_j e^{-\mathbf{r}_d^T B_j^d \mathbf{r}_d}. \end{aligned} \quad (3.55)$$

The S-wave angular momentum state is $\uparrow\uparrow$ as used previously, and the D-wave state is

$$\mathcal{Y}_{121}^1 = \sqrt{\frac{3}{5}} r_{np}^2 Y_{22} \downarrow\downarrow - \sqrt{\frac{3}{10}} r_{np}^2 Y_{21} \frac{\uparrow\downarrow + \downarrow\uparrow}{\sqrt{2}} + \sqrt{\frac{1}{10}} r_{np}^2 Y_{20} \uparrow\uparrow. \quad (3.56)$$

Using the cartesian representation of the spherical harmonics is advantageous as we can then directly apply the formulae for matrix elements we have used earlier. In cartesian coordinates[8],

$$\begin{aligned} r_{np}^2 Y_{22} &= \sqrt{\frac{15}{32\pi}} (r_{np,x} + i r_{np,y})^2 = \sqrt{\frac{15}{32\pi}} (\vec{e}_+ \cdot \vec{r}_{np})^2, \\ r_{np}^2 Y_{21} &= -\sqrt{\frac{15}{8\pi}} (r_{np,x} + i r_{np,y}) r_{np,z} = -\sqrt{\frac{15}{8\pi}} (\vec{e}_+ \cdot \vec{r}_{np}) (\vec{e}_0 \cdot \vec{r}_{np}), \\ r_{np}^2 Y_{20} &= \sqrt{\frac{15}{32\pi}} (2r_{np,z}^2 - r_{np,x}^2 - r_{np,y}^2) = \sqrt{\frac{5}{16\pi}} \left((\vec{e}_0 \cdot \vec{r}_{np})^2 - (\vec{e}_+ \cdot \vec{r}_{np}) (\vec{e}_- \cdot \vec{r}_{np}) \right), \end{aligned} \quad (3.57)$$

where $\vec{e}_+ = (1, i, 0)$, $\vec{e}_- = (1, -i, 0)$ and $\vec{e}_0 = (0, 0, 1)$. The total D-wave wave function in the relative Jacobi coordinates can then be written as

$$\begin{aligned} \psi_{np}^{(D\text{-wave})} &= \\ &3(\mathbf{a}^T \mathbf{r}_d)(\mathbf{a}^T \mathbf{r}_d) \downarrow\downarrow + 3(\mathbf{a}^T \mathbf{r}_d)(\mathbf{c}^T \mathbf{r}_d)(\uparrow\downarrow + \downarrow\uparrow) + \left(2(\mathbf{c}^T \mathbf{r}_d)(\mathbf{c}^T \mathbf{r}_d) - (\mathbf{a}^T \mathbf{r}_d)(\mathbf{b}^T \mathbf{r}_d) \right) \uparrow\uparrow \end{aligned} \quad (3.58)$$

where $\mathbf{a} = (\vec{e}_+)$, $\mathbf{b} = (\vec{e}_-)$, and $\mathbf{c} = (\vec{e}_0)$. An overall factor of $1/\sqrt{32\pi}$ has been excluded, as it will just be absorbed into the c_j -coefficients.

Because A_i^d and B_j^d are two different sets of Gaussians, the two-component wave

function Ψ is numerically treated as a three-component wavefunction (although it is still a two-component system)

$$\Psi \longrightarrow \begin{pmatrix} \psi_{np}^{(\text{S-wave})} \\ \psi_{np}^{(\text{D-wave})} \\ \psi_{NN\pi} \end{pmatrix}, \quad (3.59)$$

and the overlap matrix $\mathcal{N}$ becomes a 3x3 block matrix. Only the diagonal blocks in $\mathcal{N}$ are non-zero, as there is no overlap between $\psi_{np}^{(\text{S-wave})}$ and $\psi_{np}^{(\text{D-wave})}$. Because the spin states are orthogonal, only the last term of (3.58) could contribute to the overlap, but it does not, as

$$\langle A_i^d | B_j^d \rangle \propto (2\mathbf{c}^T \mathbf{c} - \mathbf{a}^T \mathbf{b})R = 0, \quad (3.60)$$

since R is a 1-dimensional matrix.

Even if we ignored the spin, there would still not be an overlap with the S-wave, because it would be proportional to $\mathbf{a}^T \mathbf{a}$ and $\mathbf{a}^T \mathbf{c}$, both of which are 0. This is good, because it means the D-wave state is a pure D-wave, and the probability of being in this state can then easily be found by the c_j -coefficients and the corresponding block in $\mathcal{N}$,

$$P_D = c_{\text{D-wave}}^T \mathcal{N}_{22} c_{\text{D-wave}} \quad (3.61)$$

The Hamiltonian with the inclusion of D-waves takes the form

$$H = \begin{pmatrix} \hat{K}_d & 0 & W^\dagger \\ 0 & \hat{K}_d & W^\dagger \\ W & W & \hat{K}_\pi + m_\pi \end{pmatrix}. \quad (3.62)$$

Building the D-wave via optimization

As an experiment, I also try to build the D-wave using optimization of the orbital angular momentum vectors. The D-wave is constructed as

$$\begin{aligned} \psi_{np}^{(\text{D-wave})} = & (\mathbf{v}_1^T \mathbf{r}_d)(\mathbf{v}_2^T \mathbf{r}_d) \downarrow\downarrow \sum_{i=1}^{n_{22}} c_i e^{-\mathbf{r}_d^T B_i^{22} \mathbf{r}_d} \\ & + (\mathbf{v}_3^T \mathbf{r}_d)(\mathbf{v}_4^T \mathbf{r}_d)(\uparrow\downarrow + \downarrow\uparrow) \sum_{i=1}^{n_{21}} c_i e^{-\mathbf{r}_d^T B_i^{21} \mathbf{r}_d} \\ & + \left((\mathbf{v}_5^T \mathbf{r}_d)(\mathbf{v}_6^T \mathbf{r}_d) + (\mathbf{v}_7^T \mathbf{r}_d)(\mathbf{v}_8^T \mathbf{r}_d) \right) \uparrow\uparrow \sum_{i=1}^{n_{20}} c_i e^{-\mathbf{r}_d^T B_i^{20} \mathbf{r}_d} \end{aligned} \quad (3.63)$$

where all the $\mathbf{v}_i$'s contain complex vectors, and are variational parameters, eg.

$$\mathbf{v}_1 = \begin{pmatrix} x_1 + iu_1 \\ y_1 + iv_1 \\ z_1 + iw_1 \end{pmatrix}, \quad (3.64)$$

where x, y, z are the real part of the vector, and u, v, w the imaginary part. The only constraint on the vectors is that $\mathbf{v}_5^\top \mathbf{v}_6 + \mathbf{v}_7^\top \mathbf{v}_8 = 0$, so the overlap with the S-wave is 0. Further constraints could be imposed to significantly reduce the number of parameters, eg. we could require that all the vectors be unit vectors, or that $\mathbf{v}_1^\top \mathbf{v}_2 = \mathbf{v}_3^\top \mathbf{v}_4 = 0$. The result of this experiment is best illustrated by visualizing the spherical harmonics in 3D, as seen in figure 3.1. After optimizing the randomly chosen vectors, they resemble the correct spherical harmonics.

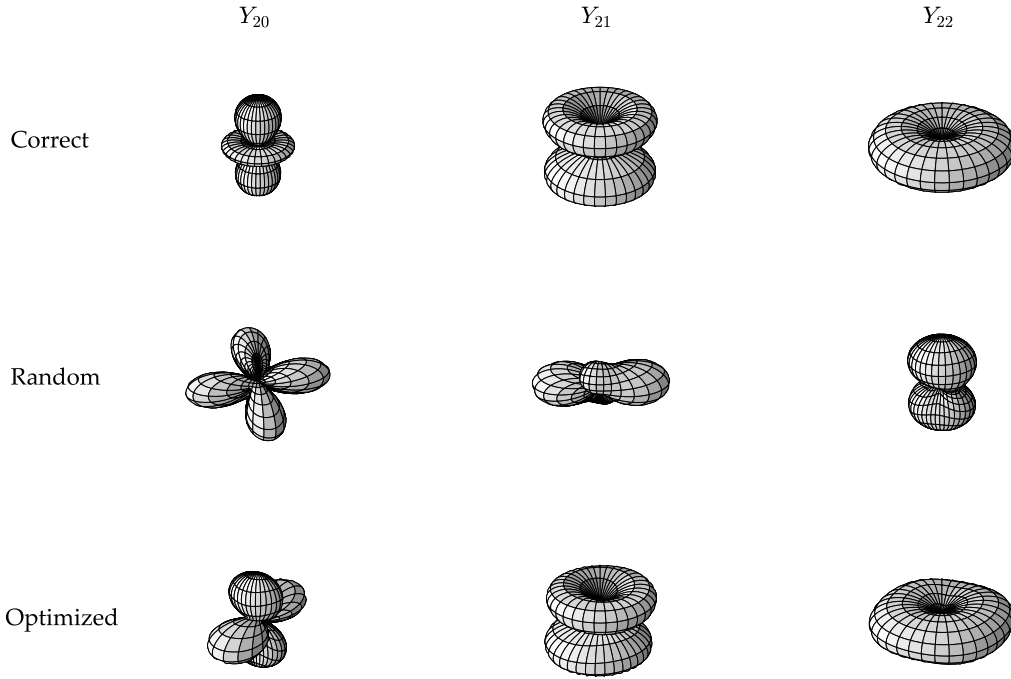


Figure 3.1: Top: Correct spherical harmonics. Middle: random guess used for the optimization. Bottom: After the optimization..

Charge radius

The D-waves also contribute to the deuteron charge radius. The contribution will be

$$R_{c,D\text{-wave}}^2 = \langle \psi_{np}^{(D\text{-wave})} | r_p^2 | \psi_{np}^{(D\text{-wave})} \rangle \quad (3.65)$$

We will use that $r_p^2 = r_p^+ r_p^- + r_z^2$ and that $\vec{r}_p = w_p \mathbf{r}_d = \frac{1}{2} \vec{r}_{np}$ ³. We can then write

$$r_p^2 = \frac{1}{4} (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) + \frac{1}{4} (\mathbf{c}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d), \quad (3.66)$$

³ The transformed w_p is equal to 1/2. Alternatively, one can think of the proton vector being half the relative vector in the center of mass system, as the nucleons have the same mass.

and the charge radius becomes

$$\begin{aligned}
 \langle \psi_{np}^{(D\text{-wave})} | r_p^2 | \psi_{np}^{(D\text{-wave})} \rangle = & \frac{1}{4} \sum_{i,j=1}^{n_D} c_i c_j \langle B_i^d | 9(\mathbf{a}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{a}^\top \mathbf{r}_d) \\
 & + 18(\mathbf{c}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d) (\mathbf{a}^\top \mathbf{r}_d) \\
 & + (\mathbf{a}^\top \mathbf{r}_d)^* (\mathbf{b}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) \\
 & + 4(\mathbf{c}^\top \mathbf{r}_d)^* (\mathbf{c}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d) \\
 & - 4(\mathbf{a}^\top \mathbf{r}_d)^* (\mathbf{b}^\top \mathbf{r}_d)^* (\mathbf{a}^\top \mathbf{r}_d) (\mathbf{b}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d) (\mathbf{c}^\top \mathbf{r}_d) \\
 & + \dots | B_j^d \rangle,
 \end{aligned} \tag{3.67}$$

where $\dots$ indicates the same 5 terms but with the middle $(\mathbf{a}^\top \mathbf{r}_d)(\mathbf{b}^\top \mathbf{r}_d)$ replaced by $(\mathbf{c}^\top \mathbf{r}_d)(\mathbf{c}^\top \mathbf{r}_d)$.

There will be no contribution from $\langle \psi_{np}^{(D\text{-wave})} | r_p^2 | \psi_{np}^{(S\text{-wave})} \rangle$, as it is 0. This can be shown by explicitly writing up the matrix element and applying the P-wave overlap formula. Then all the terms will cancel. This will not be done here though.

Quadrupole moment

The wave function not being a pure S-wave induces an electric quadrupole moment in the deuteron. The quadrupole moment is found as the expectation value[1, 18]

$$Q_D = \sum_{k=1}^N q_k \langle \hat{Q}_k \rangle = \sum_{k=1}^N q_k \langle 2z_k^2 - x_k^2 - y_k^2 \rangle, \tag{3.68}$$

where q_k is the charge of the k 'th particle, and x_k, y_k and z_k are the components of the k 'th particle position vector. Only the proton contributes to the quadrupole moment. The expectation value can be rewritten to

$$\langle 2z_p^2 - x_p^2 - y_p^2 \rangle = \frac{1}{4} \langle 2(\mathbf{c}^\top \mathbf{r})(\mathbf{c}^\top \mathbf{r}) - (\mathbf{a}^\top \mathbf{r})(\mathbf{b}^\top \mathbf{r}) \rangle. \tag{3.69}$$

The quadrupole moment is then given as

$$Q_D = 2e \langle \psi_{np}^{(S\text{-wave})} | \hat{Q}_p | \psi_{np}^{(D\text{-wave})} \rangle + e \langle \psi_{np}^{(D\text{-wave})} | \hat{Q}_p | \psi_{np}^{(D\text{-wave})} \rangle + e \langle \psi_{NN\pi} | \hat{Q}_p | \psi_{NN\pi} \rangle. \tag{3.70}$$

There is no contribution between two S-waves as these are spherically symmetric. It is also easily seen that the expectation value is indeed 0, as the equation is the same as (3.60), up to a factor.

The first term in eq. (3.70) is evaluated using the formula for the overlap of two D-waves (2.24). The middle is found as the overlap between two F-waves (2.25). The last term has the same form as eq. (3.53), where the r_k^2 will be replaced by $\hat{Q}_p$ and is then evaluated as the overlap between two P-waves (2.22).

CHAPTER 4

Results

The parameters are chosen such that the model can reproduce the accepted values of the binding energy and charge radius. These are[19]

$$E_B = -2.22 \text{ MeV} \quad R_C = 2.13 \text{ fm.} \quad (4.1)$$

These quantities are rounded to two digits, as the model serves as a proof of concept, and the objective of is not to get an accurate result down to the last known digit. The particle masses are also chosen as approximate values with $m_p = m_n = 939 \text{ MeV}$ and $m_\pi = 140 \text{ MeV}$. In reality, the nucleon masses differ by around 0.1% while the difference between the $\pi^\pm$ and π^0 is about 3%. The impact of this is expected to be small, and I will show this.

The π -system is a superposition of $np\pi^0$, $nn\pi^-$ and $pp\pi^+$, and the last state will be subjected to a Coulomb potential. I will also show that this effect can be neglected. In the rest of the project, I therefore assume the pions to have the same mass, and ignore the Coulomb effects.

Initially, I include the nucleon-nucleon confinement term in the coupling operator, $\exp(-r_{np}^2/b_{NN}^2)$ to easier find good model parameters. The binding energy of the deuteron will be the lowest energy of this system, as the energy will be 0 when the nucleons are far apart.

I then remove this confinement by slowly increasing the value of b_{NN} until it no longer changes the energy. The nucleons will now be far apart in the first excited state, which also has negative energy. The binding energy is then the energy difference between this excited state and the ground state. Because the pion is still confined to one of the nucleons, the energy of the first excited state is the binding energy between the virtual pion and one of the nucleons. I compare this value to a model of a dressed nucleon with the same parameters.

4.1 Limiting the inter-nucleon range

With a limit on the inter-nucleon range, the model has three parameters, b_{NN} , b_w and S_w , and only needs to fit two quantities. The variational parameter b_{ij} ¹ is stochastically sampled using an exponential distribution with a scale of b_{NN} ,

$$b_{ij} = -\ln(u)b_{\text{NN}} \quad (4.2)$$

where u is a randomly chosen number between 0 and 1. When chosen quasi-randomly, Van der Corput sequences with different prime bases for each ij set are used. Figure 4.1 shows that scaling the Gaussians by b_{NN} leads to the lowest binding energy.

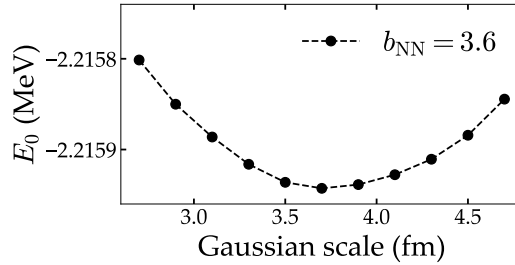


Figure 4.1: The binding energy as a function of Gaussian scale. For each point, 25 Gaussians in the np-system and 300 in the π -system were used. The model parameters are $b_{\text{NN}} = 3.6$ fm, $b_w = 1.5$ fm, $S_w = 75.3$ MeV.

The nucleon size is about 0.8 fm and the pions are a bit smaller[20, 21]. A suitable value for b_w might then be roughly around their combined size. Table 4.1 shows three sets of parameters that give the correct energy and charge radius.

b_{NN} (fm)	b_w (fm)	S_w (MeV)	Norm $_{\pi}$ (%)
3.6	1.50	75.3	2.2
3.8	1.25	85.5	2.0
4.1	1.00	102.9	1.7

Table 4.1: The sets of model parameters giving the correct binding energy and charge radius with the contribution of the π -system to the total norm.

The first set of parameters will be used in the subsequent figures in this section.

¹ As a reminder, the Gaussians are chosen as

$$\psi = \exp \left(- \sum_{i < j=1}^N \left(\frac{\vec{r}_i - \vec{r}_j}{b_{ij}} \right)^2 \right),$$

The model uses 25 Gaussians in the np-system and 300 in the π -system. This is more than enough for the energy to converge, as seen on figure 4.2. I compare both pseudo and quasi-random sampling to see that both converge to the same value. Optimization of the Gaussians also gives the same result.

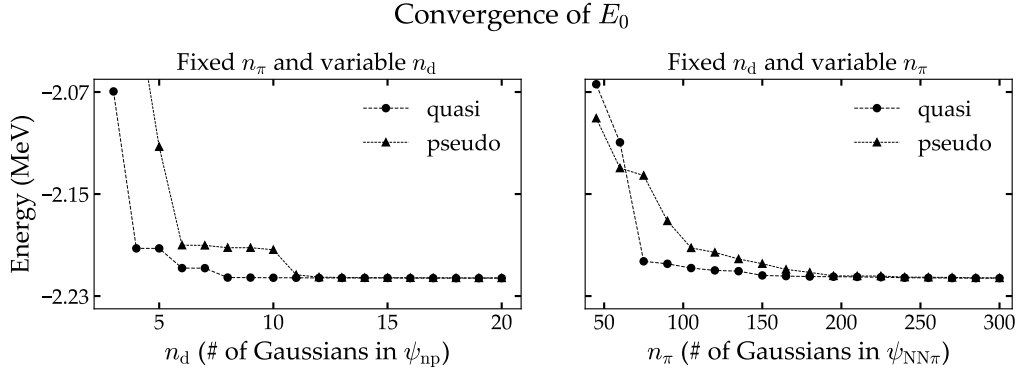


Figure 4.2: Convergence of the ground state. Left: Convergence for an increasing number of Gaussians in the np-system with 300 Gaussians in the π -system. Right: The other way around, with 25 Gaussians in the np-subsystem.

To get an idea of what combinations of parameters yield the correct ground state energy and charge radius, I have analyzed the parameter space as seen on figure 4.3. It seems for a fixed value of $b_{NN} = 3.6$ fm, the charge radius is very close to 2.13 fm if the ground state energy is -2.22 MeV. For $b_w = 1.5$ fm and $S_w = 75.3$ MeV, the two lines intersect, as indicated by the star.

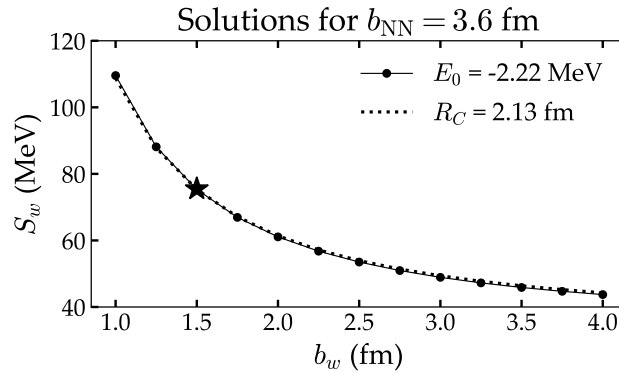


Figure 4.3: Values of b_w and S_w that give the correct ground state (solid line with points) and charge radius (dotted line). The intersection point is marked by a star.

Both lines nearly overlap, and for the small values of S_w , the charge radius is about 3% too big. For the large values, it is about 3% too small.

The radial wave function for the bare neutron-proton state is shown on figure 4.4. Its asymptotic form $e^{-\kappa r}$ with $\kappa = \sqrt{2\mu_{np}/\hbar^2}$ is also graphed on the figure.

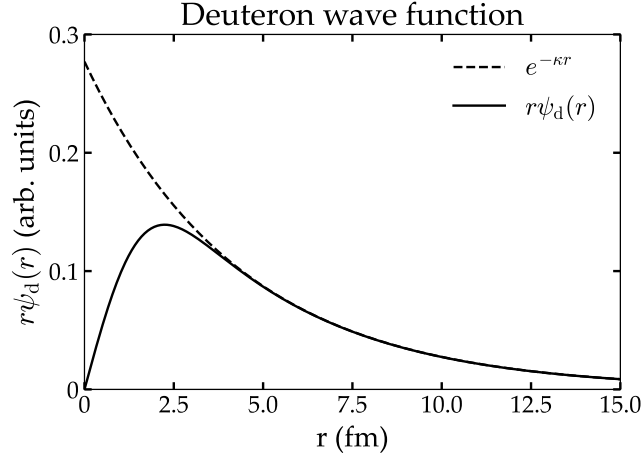


Figure 4.4: The radial wave function $r\psi_d(r)$ of the relative distance between the nucleons. The dashed line is its asymptotic form..

4.2 Different pion masses and Coulomb effects

The wavefunction $\psi_{NN\pi}$ contains a factor

$$\left((\vec{\tau}_1 \vec{\pi})(\vec{\sigma}_1 \vec{r}_{\pi 1}) + (\vec{\tau}_2 \vec{\pi})(\vec{\sigma}_2 \vec{r}_{\pi 2}) \right). \quad (4.3)$$

where the $(\vec{\tau} \vec{\pi})$ products can be expanded into three terms,

$$(\vec{\tau} \vec{\pi}) = \tau_0 \pi^0 + \sqrt{2} \tau_- \pi^+ + \sqrt{2} \tau_+ \pi^-. \quad (4.4)$$

When applying the τ operators on the isospin state, we see that the wave function is a superposition of an $np\pi^0$ state, an $nn\pi^+$ state and a $pp\pi^-$ state. The last one is affected by a Coulomb potential. To include this effect, the three states in the π -system get different sets of Gaussians. Because the three pions are orthogonal to each other, the Hamiltonian and overlap matrices will become 4×4 block matrices, and the two-component wavefunction

$$\Psi = \begin{pmatrix} \psi_{np} \\ \psi_{NN\pi} \end{pmatrix}, \quad (4.5)$$

will effectively be treated as a four-component wavefunction,

$$\Psi \rightarrow \begin{pmatrix} \psi_{np} \\ \psi_{np\pi^0} \\ \psi_{nn\pi^+} \\ \psi_{pp\pi^-} \end{pmatrix}, \quad (4.6)$$

although in reality,

$$\psi_{NN\pi} = \psi_{np\pi^0} + \psi_{nn\pi^+} + \psi_{pp\pi^-}. \quad (4.7)$$

The Hamiltonian then looks like

$$H = \begin{pmatrix} \hat{K}_d & W_{\pi^0}^\dagger & W_{\pi^+}^\dagger & W_{\pi^-}^\dagger \\ W_{\pi^0} & \hat{K}_{\pi^0} + m_{\pi^0} & 0 & 0 \\ W_{\pi^+} & 0 & \hat{K}_{\pi^+} + m_{\pi^+} & 0 \\ W_{\pi^-} & 0 & 0 & \hat{K}_{\pi^-} + m_{\pi^-} + V_C \end{pmatrix}, \quad (4.8)$$

where the matrices in the kinetic energy operators for the different pions are

$$K_{\pi^\alpha} = \begin{pmatrix} K_{np} & 0 \\ 0 & \frac{\hbar^2}{2\mu_{NN'\pi^\alpha}} \end{pmatrix}. \quad (4.9)$$

The W operator has also been split into three components, such that W_{π^0} generates a π^0 and so on.

Having split the π -system into three states also allows the pions to have different masses. I use $m_{\pi^\pm} = 140$ MeV and $m_{\pi^0} = 135$ MeV. The nucleons have the same mass in this model, so the reduced masses $\mu_{NN'\pi^\alpha}$ are only different due to the pion masses.

The matrix elements for the three π states are the same as earlier, but a factor 3 smaller, because we do not get the 3 from $(\vec{\tau}\vec{\pi})^\dagger(\vec{\tau}\vec{\pi})$ anymore. In addition, the matrix element for the $pp\pi^-$ state gets three extra terms from the Coulomb potential, because there is the proton-proton repulsion and the proton-pion attraction for each proton.

	Different π masses	Coulomb	total
E_{changed}/E_0 (%)	97.6	100.3	97.9

Table 4.2: Difference in the ground state energy depending on the changes to the model.

Table 4.2 shows the effect of using different pion masses and adding a Coulomb potential to the Hamiltonian. Having different masses reduces the ground state energy, while the Coulomb potential increases it. The total effect is therefore less

than if we only changed the masses.

Adding the Coulomb potential amounts to a difference of 0.006 MeV, which is smaller than the accuracy of the binding energy (0.01 MeV). Nevertheless, I check this result with the expectation value of the Coulomb potential, $\langle V_C \rangle$, and it is the same.

This experiment further shows that the pions do not contribute to the charge radius, as the contribution from the Coulomb potential is so small, it only has a negligent effect on the charge radius. Likewise, all protons contribute the same, and in total, they add up to the value as one of the nucleons in the π -system.

4.3 Dressing of a nucleon

Removing b_{NN} allows the model to produce states where the nucleons are far apart and have negative energy. To see the effect, I slowly increase b_{NN} until it no longer decreases the energy, as shown on figure 4.5.

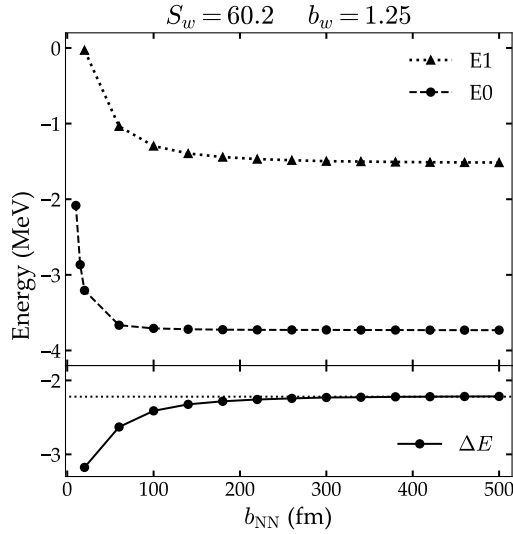


Figure 4.5: Top: Convergence of the two lowest energy levels as a function of the range parameter b_{NN} . The other model parameters are $b_w = 1.25$ fm and $S_w = 60.2$ MeV. Bottom: Difference in energy.

During this process, the pion will stick to one of the bare nucleons, dressing it. This is the first excited state of the system and its energy corresponds to the binding energy between the virtual pion and the nucleon. It will also become the threshold for the bound state. As such, the deuteron binding energy will be the difference between the two lowest energies. Using stochastic sampling, the

model had a hard time getting the excited state to converge. Hence, all the results are produced by minimization, where each of the energy levels is granted a set of Gaussians to be optimized. The np-system is given 7 Gaussians to each level, and the π -system is given 30 to the ground state and 60 to the excited state. The ground state was found to converge easier and as such required fewer Gaussians.

When b_{NN} is removed, the model parameters have to be retuned to the binding energy. For $b_w = 1.25$ fm and $b_w = 1.5$ fm, the binding energy as a function of S_w can be seen on figure 4.6

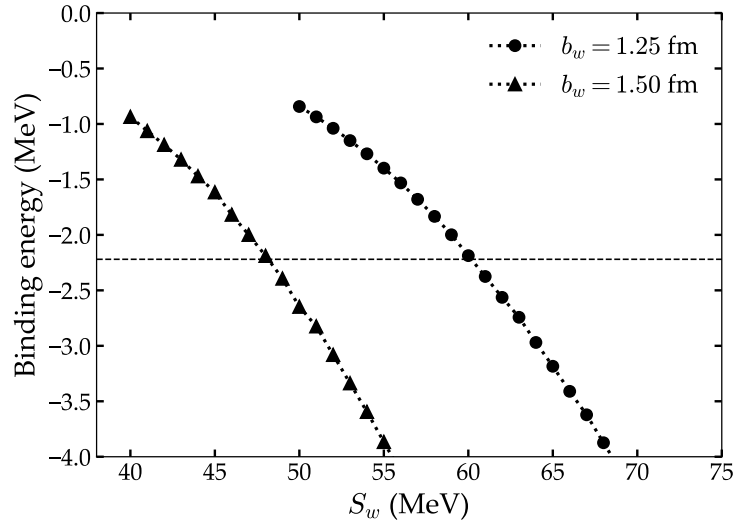


Figure 4.6: Binding energy of the deuteron as a function of the coupling strength S_w for two values of b_w . E_1 is very close to 0 at the first point on each graph.

As listed in table 4.3, when $b_w = 1.5$ fm, the charge radius is a bit smaller than 3 fm, and for $b_w = 1.25$, it is a bit less. Even for smaller values of the range b_w , the model cannot produce a charge radius of 2.13 fm. The table also shows the binding energy of a bare nucleon dressed with a virtual pion. In appendix B I make a model of the nucleon in one-pion MEM to compare the binding energy of the virtual pion. Using the same parameters, the single nucleon model produces a binding energy of -16.2 MeV for $b_w = 1.5$ fm and -20.1 MeV for $b_w = 1.25$ fm.

b_w (fm)	S_w (MeV)	E_π (MeV)	R_C (fm)	Norm $_\pi$ (%)
1.50	48.2	-1.18	2.98	2.0
1.25	60.2	-1.52	2.87	1.9
1.00	79.5	-1.83	2.85	1.7

Table 4.3: Model parameters giving the correct binding energy. E_π is the binding energy between a bare nucleon and a pion. The ground state energy is 2.22 MeV lower than E_π .

4.4 Deuteron with D-waves

I include D-waves in the wave function to hopefully allow the model to produce the accepted values for both the binding energy and the charge radius. Further, the addition of D-waves gives the deuteron a quadrupole moment that can be calculated. The experimental value of this is[19]

$$Q_D = 0.29 \text{ e} \cdot \text{fm}^2 \quad (4.10)$$

Sometimes the quadrupole moment is written in units of e , as the charge is also present in the equation of Q_D (3.68).

No nucleon dressing

At first, b_{NN} is set to a small value such that the nucleons are not allowed to be dressed by themselves. Table 4.4 shows the results from this model. The same sets of parameters as earlier are used to see the effect the addition of D-waves has on the system. Another set of parameters is fit to the binding energy, charge radius, and quadrupole moment.

b_{NN} (fm)	b_w (fm)	S_w (MeV)	E_0 (MeV)	R_C (fm)	Q_D (fm ²)	N_D	N_π
3.6	1.50	75.3	-2.42	2.07	0.14	0.5	2.4
3.8	1.25	85.5	-2.51	2.04	0.16	0.7	2.1
4.1	1.00	102.9	-2.66	2.01	0.19	1.0	1.9
4.4	0.65	159	-2.22	2.13	0.29	1.8	1.2

Table 4.4: Same parameters as for the S-wave system, and another set, tuned to the binding energy, charge radius, and quadrupole moment. The N_D and N_π columns are the contribution to the total norm of the D-wave state and the π -state in %. The units have been omitted to reduce the width of the table.

Allowing dressing

b_{NN} is again removed, and the threshold is moved down. The addition of D-waves has a significant impact on the model. The coupling strength needs to be increased a lot from the results without D-waves, to still produce the correct binding energy, as shown on figure 4.7. The energy curves are no longer monotone, yet they still only cross the binding energy line once.

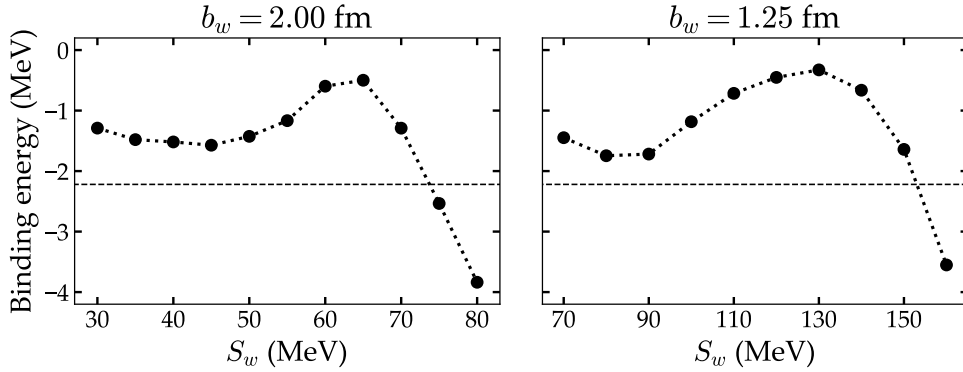


Figure 4.7: Binding energy as a function of coupling strength S_w for two different values of b_w . The figure is made for illustrative purposes, and the actual energy may differ by a few percent.

Only one set of parameters was found that produces the correct binding energy and charge radius. These are $b_w = 1.94$ fm and $S_w = 80.2$ MeV. The addition of D-waves pushes the energy levels down a lot. The nucleon-pion binding energy for these parameters is -35.6 MeV. For the same parameters, the single nucleon model produces a binding energy of -52.4 MeV.

Calculating the quadrupole moment gives 0.64 fm². The contribution from the pure D-state is expected to be negative[19], and is -0.31 fm², while the mixing of the S- and D-state is 0.93 fm². There is also a very small contribution from the π -state of about 0.01 fm². The D-waves take up 19% of the total norm, while the π -state only takes up 12%.

Discussion

Initially, with three model parameters, reasonable values can be chosen that produce the binding energy and charge radius of the deuteron. This was expected because the three parameters were fitted to two values. When including the D-waves, the nucleon-pion range has to be chosen as 0.65 fm to also accommodate the experimental value of the quadrupole moment. This seems too small, as it is smaller than the nucleon.

Without the D-waves, when increasing b_{NN} and allowing dressing, we see the new energy levels converge around a few hundred fm. The energies do not move down a lot. This means the binding energy of the pion is only a few MeV. This is a lot smaller than the value predicted by the single nucleon model of around -20 MeV. The model also cannot produce the deuteron charge radius. It seems that the S-wave, one-pion approximation is not accurate enough to describe the deuteron. The next step would be to either add another pion or include the D-waves to the model, knowing that the deuteron is not a pure S-wave. The addition of D-waves allows the model to also produce the charge radius. This is achieved using $b_w = 1.94$ fm and $S_w = 80.2$ MeV. The value of the range parameter may be a little larger than expected, as the combined radius of a nucleon and a pion is around 1.5 fm. However, the D-waves also have a significant impact on the energy levels of the system, pushing the threshold energy down to -35.6 MeV for these parameters. For the same parameters, the single nucleon model still produces a larger binding energy, yet with the inclusion of D-waves, the relative difference is reduced.

The model does not produce the correct quadrupole moment, by a factor of about 2. One might be inclined to think that it could be due to a rouge factor 2 in the expression for the quadrupole expectation value, however, taking up around 4 times as much of the norm as other deuteron models, suggests that this may not

be the case.

The next natural step would be the inclusion of another pion. Because the binding energy is calculated as the difference between the two lowest energy states, and another pion would push the threshold down more than a single pion, this may improve the results.

It may also be necessary to investigate the effects of mass renormalization. The bare mass of the nucleons should be used in place of m_n and m_p . This is done by first using the physical mass of the nucleons, running the model, and subtracting the nucleon-pion binding energy from the mass. This is done a few times until the energy converges. I have not included this, as I expect it to only be a small correction since the binding energy was predicted to only be a few % of the nucleon masses.

Another aspect of improvement would be on the optimization side of the model. When b_{NN} is small, minimization takes around 30 seconds. Removing b_{NN} , minimization can take anywhere from 20 minutes to an hour when D-waves are included because two levels need to converge. Because of the long run times, it is very advantageous to reuse optimized Gaussians when the model parameters change slightly. These will be optimized much faster than a random guess.

When this is not an option, another strategy would be to make a better guess. A quick analysis showed that using the same scale parameter on the Gaussians in the three-body state is not ideal, however, a more thorough investigation of what Gaussians gives a good energy is needed to improve the guess.

Conclusion

I describe the deuteron in the nuclear model with explicit mesons. As a first approximation, I only include one pion. The model relies on the variational principle, utilizing explicitly correlated Gaussians as basis functions. The Gaussians benefits from having analytic matrix elements which can greatly reduce the computation time. With this formalism, orbital angular momentum wave functions are easily added to the system and their matrix elements are also analytic.

With a limit on the nucleon-nucleon range, the model easily reproduces the ground state energy and charge radius of the deuteron. One of the features of the model is its ability to also describe dressed nucleons. To do this, we cannot limit the nucleon-nucleon range. Ideally, the model should be able to predict the binding energy between a virtual nucleon. Comparing the deuteron to a single nucleon described in one pion MEM shows that this does not seem to be the case. As the model also cannot produce the accepted value of the charge radius, it is modified to include D-waves. This allows it to reproduce the charge radius, as well as increase the binding energy of the pion, moving it closer to the value estimated by the single nucleon model. With the description of D-waves, I calculate the quadrupole moment, which turns out about twice as big as the experimental value. There is also too much D-wave in the deuteron norm, compared to different deuteron descriptions.

It seems further work needs to be done. Perhaps the two-pion approximation will suffice.

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Derivation of matrix elements

The P- and D-wave matrix elements can be derived by differentiating The S-wave elements with respect to the shift vectors. We will use $\partial_{\mathbf{v}^\top}$ as shorthand notation for the partial derivative w.r.t. $\mathbf{v}^\top$. To apply the derivatives, we write out the vector products in indices. Whenever the same index appears twice, it is assumed to be summed over. Indices on the bold vectors will be denoted in the subscript by characters from the english alphabet. The 3D vectors will be denoted by greek characters in the superscript.

Some commonly used relations we need, are

$$(\vec{a} \cdot \partial_{\vec{v}}) \vec{v} = \vec{a} \quad (\text{A.1})$$

$$(\vec{a} \cdot \partial_{\vec{v}})(\vec{b} \cdot \vec{v}) = \left(\vec{b} \cdot [(\vec{a} \cdot \partial_{\vec{v}}) \vec{v}] \right) = \vec{b} \cdot \vec{a} \quad (\text{A.2})$$

$$(\mathbf{a}^\top \partial_{\mathbf{v}}^\top)(\mathbf{b}^\top \mathbf{v}) = (\vec{a}_i \cdot \partial_{\vec{v}_i})(\vec{b}_j \cdot \vec{v}_j) = (\vec{b}_j \cdot \vec{a}_i) \delta_{ij} = (\mathbf{b}^\top \mathbf{a}) \quad (\text{A.3})$$

$$\partial_{\mathbf{v}}^\top (\mathbf{v}^\top R \mathbf{v}) = (R \mathbf{v} + R^\top \mathbf{v}). \quad (\text{A.4})$$

The last one can be shown by considering its i 'th component,

$$\begin{aligned} \left(\partial_{\mathbf{v}}^\top (\mathbf{v}^\top R \mathbf{v}) \right)_i &= \partial_{v_i} (\vec{v}_n \cdot \vec{v}_m) R_{nm} \\ &= (\delta_{ni} \vec{v}_m R_{nm} + \vec{v}_n \delta_{mi} R_{nm}) \\ &= (R \mathbf{v} + R^\top \mathbf{v})_i. \end{aligned} \quad (\text{A.5})$$

It then follows that

$$(\mathbf{a}^\top \partial_{\mathbf{v}}^\top)(\mathbf{v}^\top R \mathbf{v}) = \mathbf{a}^\top (R + R^\top) \mathbf{v} \quad (\text{A.6})$$

Overlap

S-waves

The overlap between two shifted S-waves with parameters B and A , and with shift vectors $\mathbf{t}$ and $\mathbf{s}$, is

$$\langle B\mathbf{t}|A\mathbf{s}\rangle = e^{\frac{1}{4}\mathbf{v}^\top R\mathbf{v}} \left(\frac{\pi^{\dim(A)}}{\det(A+B)} \right)^{3/2} \equiv M, \quad (\text{A.7})$$

where $\mathbf{v} = \mathbf{s} + \mathbf{t}$ and $R = (A+B)^{-1}$. In the limit where $\mathbf{v} \rightarrow 0$, $M \rightarrow M_0$.

P-waves

The overlap between two shifted P-wave gaussians is

$$\langle B\mathbf{t}|(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\mathbf{s}\rangle = \frac{1}{2}\mathbf{b}^\top R\mathbf{a}M + \frac{1}{4}(\mathbf{v}^\top R\mathbf{b})(\mathbf{v}^\top R\mathbf{a})M. \quad (\text{A.8})$$

To show this, we use

$$\langle B\mathbf{t}|(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\mathbf{s}\rangle = (\mathbf{b}^\top \partial_{\mathbf{v}\tau})(\mathbf{a}^\top \partial_{\mathbf{v}\tau})\langle B\mathbf{t}|A\mathbf{s}\rangle, \quad (\text{A.9})$$

and

$$(\mathbf{a}^\top \partial_{\mathbf{v}\tau})e^{\frac{1}{4}\mathbf{v}^\top R\mathbf{v}} = e^{\frac{1}{4}\mathbf{v}^\top R\mathbf{v}} \frac{1}{4}(\mathbf{a}^\top \partial_{\mathbf{v}\tau})(\mathbf{v}^\top R\mathbf{v}) = e^{\frac{1}{4}\mathbf{v}^\top R\mathbf{v}} \frac{1}{2}(\mathbf{v}^\top R\mathbf{a}), \quad (\text{A.10})$$

because R is symmetric.

Eq. (A.8) then follows directly when differentiating again. The unshifted overlap shall be denoted M_1 ,

$$M_1 \equiv \frac{1}{2}\mathbf{b}^\top R\mathbf{a}M_0. \quad (\text{A.11})$$

D-waves

The overlap between two shifted D-wave gaussians is found by carrying out another two differentiations of eq. (A.8), and is

$$\begin{aligned} \langle B\mathbf{t}|(\mathbf{d}^\top \mathbf{r})(\mathbf{c}^\top \mathbf{r})(\mathbf{b}^\top \mathbf{r})(\mathbf{a}^\top \mathbf{r})|A\mathbf{s}\rangle = & \\ & \frac{1}{4}\left\{(\mathbf{d}^\top R\mathbf{a})(\mathbf{c}^\top R\mathbf{b}) + (\mathbf{d}^\top R\mathbf{b})(\mathbf{c}^\top R\mathbf{a}) + (\mathbf{d}^\top R\mathbf{c})(\mathbf{b}^\top R\mathbf{a})\right\}M \\ & + \frac{1}{8}\left\{(\mathbf{d}^\top R\mathbf{a})(\mathbf{v}^\top R\mathbf{b})(\mathbf{v}^\top R\mathbf{c}) + (\mathbf{v}^\top R\mathbf{a})(\mathbf{d}^\top R\mathbf{b})(\mathbf{v}^\top R\mathbf{c}) + (\mathbf{v}^\top R\mathbf{a})(\mathbf{v}^\top R\mathbf{b})(\mathbf{d}^\top R\mathbf{c})\right\}M \\ & + \frac{1}{8}\left\{(\mathbf{v}^\top R\mathbf{a})(\mathbf{c}^\top R\mathbf{b}) + (\mathbf{v}^\top R\mathbf{b})(\mathbf{c}^\top R\mathbf{a}) + (\mathbf{b}^\top R\mathbf{a})(\mathbf{v}^\top R\mathbf{c}) + \frac{1}{2}(\mathbf{v}^\top R\mathbf{a})(\mathbf{v}^\top R\mathbf{b})(\mathbf{v}^\top R\mathbf{c})\right\}M. \end{aligned} \quad (\text{A.12})$$

Only the first curly parenthesis survives when $\mathbf{v} \rightarrow 0$, and $M \rightarrow M_0$. We denote this overlap M_2 .

F-waves

For the overlap between two F-waves, I will not show the shifted overlap, and refer to eq. (2.25) for the unshifted overlap. To derive it, one can take eq. (A.12), and write up all the combinations two extra derivatives would produce. Then, use that only an even number of derivatives on M gives something non-zero when $\mathbf{v} \rightarrow 0$, and likewise discard terms where the number of $\mathbf{v}$'s is not equal to the number of derivatives.

This then results in the 15 terms of eq. (2.25).

There seems to be a trend: for every two pre-factors, we divide by 2, and the terms in the formula will be all the possible unique permutation of all the $\mathbf{a}, \mathbf{b}, \mathbf{c} \dots$ vectors. $(\mathbf{a}^\top R \mathbf{b}) = (\mathbf{b}^\top R \mathbf{a})$ as they are just numbers. For n even pre-factors, there will be $n!$ possible permutations but given that $(\mathbf{a}^\top R \mathbf{b}) = (\mathbf{b}^\top R \mathbf{a})$, we divide by 2 for every two terms. Then, we are left with $(n/2)!$ identical terms because $(\mathbf{a}^\top R \mathbf{b})(\mathbf{c}^\top R \mathbf{d}) = (\mathbf{c}^\top R \mathbf{d})(\mathbf{b}^\top R \mathbf{a})$, so the total number of terms in the expectation value of n factors is

$$\frac{n!}{2^{n/2}(n/2)!} \quad (\text{A.13})$$

Kinetic matrix elements

S-waves

With the kinetic energy operator on the form

$$\hat{K} = -\frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top}, \quad (\text{A.14})$$

the matrix element between to shifted S-waves is

$$\langle B\mathbf{t} | \hat{K} | A\mathbf{s} \rangle = \left(6\text{Tr}(BKAR) + (\mathbf{t} - BR\mathbf{v})^\top K(\mathbf{s} - AR\mathbf{v}) \right) M. \quad (\text{A.15})$$

To show this, we first establish that

$$\langle B\mathbf{t} | -\frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top} | A\mathbf{s} \rangle = \left\langle \frac{\partial}{\partial \mathbf{r}} B\mathbf{t} | K | \frac{\partial}{\partial \mathbf{r}^\top} A\mathbf{s} \right\rangle, \quad (\text{A.16})$$

ie. the left derivative will act on the $B\mathbf{t}$ Gaussian at the cost of a sign. This can be shown using integration by parts, and that the Gaussians vanish exponentially at $\pm\infty$ ¹.

¹ $-\int_{\mathbb{R}^{3n}} d^n \mathbf{r} f(\mathbf{r}) \frac{\partial}{\partial r_i} \frac{\partial}{\partial r_j} g(\mathbf{r}) = -\int_{\mathbb{R}^{3n}} d^n \mathbf{r} f(\mathbf{r}) \frac{\partial}{\partial r_i} g^{(j)}(\mathbf{r}) = -f(\mathbf{r}) g^{(j)}(\mathbf{r})|_{-\infty}^{\infty} + \int_{\mathbb{R}^{3n}} d^n \mathbf{r} f^{(i)}(\mathbf{r}) g^{(j)}(\mathbf{r})$

with $g^{(j)} = \frac{\partial}{\partial r_j} g(\mathbf{r})$

Then, using the relation

$$\frac{\partial}{\partial \mathbf{r}^\top} e^{-\mathbf{r}^\top A \mathbf{r} + \mathbf{s}^\top \mathbf{r}} = \frac{\partial}{\partial \mathbf{r}^\top} \left\{ -\mathbf{r}^\top A \mathbf{r} + \mathbf{s}^\top \mathbf{r} \right\} e^{-\mathbf{r}^\top A \mathbf{r} + \mathbf{s}^\top \mathbf{r}} = (\mathbf{s} - 2A\mathbf{r}) e^{-\mathbf{r}^\top A \mathbf{r} + \mathbf{s}^\top \mathbf{r}}, \quad (\text{A.17})$$

the matrix element can be rewritten as

$$\langle B\mathbf{t} | \hat{K} | A\mathbf{s} \rangle = \langle B\mathbf{t} | (\mathbf{t} - 2B\mathbf{r})^\top K (\mathbf{s} - 2A\mathbf{r}) | A\mathbf{s} \rangle, \quad (\text{A.18})$$

which can be evaluated by substituting the $\mathbf{r}$'s by derivatives of the shifts $\mathbf{t}$ and $\mathbf{s}$,

$$\begin{aligned} \langle B\mathbf{t} | \hat{K} | A\mathbf{s} \rangle &= \left(\mathbf{t} - 2B \frac{\partial}{\partial \mathbf{t}^\top} \right)^\top K \left(\mathbf{s} - 2A \frac{\partial}{\partial \mathbf{s}^\top} \right) \langle B\mathbf{t} | A\mathbf{s} \rangle \\ &= \left(\mathbf{t}^\top K \mathbf{s} + 4 \frac{\partial}{\partial \mathbf{t}} B K A \frac{\partial}{\partial \mathbf{s}^\top} - 2 \mathbf{t} K A \frac{\partial}{\partial \mathbf{s}^\top} - 2 \frac{\partial}{\partial \mathbf{t}} B K \mathbf{s} \right) e^{\mathbf{v}^\top R \mathbf{v} / 4} M_0, \end{aligned} \quad (\text{A.19})$$

where we can use

$$\frac{\partial}{\partial \mathbf{s}^\top} e^{\frac{1}{4} \mathbf{v}^\top R \mathbf{v}} = \frac{\partial}{\partial \mathbf{s}^\top} \left\{ \frac{1}{4} \mathbf{v}^\top R \mathbf{v} \right\} e^{\frac{1}{4} \mathbf{v}^\top R \mathbf{v}} = \frac{1}{2} R \mathbf{v} e^{\frac{1}{4} \mathbf{v}^\top R \mathbf{v}}, \quad (\text{A.20})$$

This means we can replace all single derivatives in eq. (A.19) with $R\mathbf{v}/2$.

Then we only need to evaluate the second term in eq. (A.19), for which we need

$$\frac{\partial}{\partial \mathbf{t}} F R \mathbf{v} = \sum_{ij} \frac{\partial}{\partial t_i} \cdot \vec{v}_j (F R)_{ij} = \sum_{ij} 3 \delta_{ij} (F R)_{ij} = 3 \text{Tr}(F R), \quad (\text{A.21})$$

and when we apply the product rule on the second term in eq. (A.19), we get the expectation value for the kinteic energy operator to be

$$\langle B\mathbf{t} | \hat{K} | A\mathbf{s} \rangle = \left(6 \text{Tr}(B K A R) + (\mathbf{t} - B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v}) \right) M. \quad (\text{A.22})$$

And when $\mathbf{v} \rightarrow 0$,

$$\langle B | \hat{K} | A \rangle = 6 \text{Tr}(B K A R) M_0 \equiv L M_0. \quad (\text{A.23})$$

P-waves

The expectation value of the kinetic energy operator between two shifted P-wave guassians is

$$\begin{aligned} \langle B\mathbf{t} | (\mathbf{c}^\top \mathbf{r}) \hat{K} (\mathbf{a}^\top \mathbf{r}) | A\mathbf{s} \rangle &= M \left\{ \mathbf{c}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{a} \right. \\ &\quad + (L + (\mathbf{t} - B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v})) \left(\frac{1}{2} \mathbf{c}^\top R \mathbf{a} + \frac{1}{4} (\mathbf{v}^\top R \mathbf{a}) (\mathbf{v}^\top R \mathbf{c}) \right) \\ &\quad + ((\mathbf{c} - B R \mathbf{c})^\top K (\mathbf{s} - A R \mathbf{v}) - (\mathbf{t} - B R \mathbf{c})^\top K A R \mathbf{c}) \frac{1}{2} \mathbf{v}^\top R \mathbf{a} \\ &\quad \left. + ((\mathbf{c} - B R \mathbf{c})^\top K (\mathbf{s} - A R \mathbf{v}) - \mathbf{a}^\top R B K (\mathbf{s} - A R \mathbf{s})) \frac{1}{2} \mathbf{v}^\top R \mathbf{c} \right\}. \end{aligned} \quad (\text{A.24})$$

It can be derived by replacing $(\mathbf{a}^\top \mathbf{r})$ with $(\mathbf{a}^\top \partial_{\mathbf{s}\tau})$ and $(\mathbf{c}^\top \mathbf{r})$ with $(\mathbf{c}^\top \partial_{\mathbf{t}\tau})$ and multiplying by -1 , as before. Then, applying these two derivatives to the unshifted S-wave matrix element for the kinetic energy operator,

$$\begin{aligned} \langle B\mathbf{t} | (\mathbf{c}^\top \mathbf{r}) \hat{K} (\mathbf{a}^\top \mathbf{r}) | A\mathbf{s} \rangle &= M (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) \left(L + (\mathbf{t} B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v}) \right) \\ &+ \left(L + (\mathbf{t} B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v}) \right) (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) M \\ &+ \left\{ (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) \left(L + (\mathbf{t} B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v}) \right) \right\} (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) M \\ &+ \left\{ (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) \left(L + (\mathbf{t} B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v}) \right) \right\} (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) M. \end{aligned} \quad (\text{A.25})$$

an carrying out the derivatives, gives the above result. For $\mathbf{v} \rightarrow 0$ becomes

$$\langle B | (\mathbf{b}^\top \mathbf{r}) \hat{K} (\mathbf{a}^\top \mathbf{r}) | A \rangle = L M_1 + \mathbf{c}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{a}. \quad (\text{A.26})$$

D-waves

Again, I will only present the unshifted matrix element, as that in iteself is very long. It is

$$\begin{aligned} \langle B | (\mathbf{d}^\top \mathbf{r}) (\mathbf{c}^\top \mathbf{r}) \hat{K} (\mathbf{b}^\top \mathbf{r}) (\mathbf{a}^\top \mathbf{r}) | A \rangle &= L M_2 \\ &+ \mathbf{d}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{b} \frac{1}{2} (\mathbf{a}^\top R \mathbf{c}) M_0 \\ &+ \mathbf{c}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{b} \frac{1}{2} (\mathbf{a}^\top R \mathbf{d}) M_0 \\ &+ \mathbf{d}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{a} \frac{1}{2} (\mathbf{b}^\top R \mathbf{c}) M_0 \\ &+ \mathbf{c}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{a} \frac{1}{2} (\mathbf{b}^\top R \mathbf{d}) M_0 \\ &- \mathbf{d}^\top \left((1 - R B) K A R + R A K (1 - B R) \right) \mathbf{c} \frac{1}{2} (\mathbf{a}^\top R \mathbf{b}) M_0 \\ &- \mathbf{b}^\top \left((1 - R A) K B R + R B K (1 - A R) \right) \mathbf{a} \frac{1}{2} (\mathbf{c}^\top R \mathbf{d}) M_0. \end{aligned} \quad (\text{A.27})$$

There is no nice way of writing up the derivation of this, so I will just sketch the idea. Let $\tilde{K} = (\mathbf{t} - B R \mathbf{v})^\top K (\mathbf{s} - A R \mathbf{v})$. Then the following identities are needed,

$$\begin{aligned} (\mathbf{b}^\top \partial_{\mathbf{s}\tau}) (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) \tilde{K} &= -\mathbf{b}^\top \left(R B K (1 - A R) + (1 - R A) K B R \right) \mathbf{a} \\ (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) (\mathbf{a}^\top \partial_{\mathbf{s}\tau}) \tilde{K} &= \mathbf{c}^\top \left((1 - R B) K (1 - A R) + R A K B R \right) \mathbf{a} \\ (\mathbf{d}^\top \partial_{\mathbf{t}\tau}) (\mathbf{c}^\top \partial_{\mathbf{t}\tau}) \tilde{K} &= \mathbf{d}^\top \left((1 - R B) K (1 - A R) + R A K (1 - B R) \right) \mathbf{c}. \end{aligned} \quad (\text{A.28})$$

Then, by considering $\tilde{K}$ as a single term in (A.22), we can write up all the different combinations of the four derivatives using the product rule. This gives 16 terms. We then throw away terms with an unven number of derivatives of M , as well as the term with no derivatives of M (This has four derivatives of $\tilde{K}$, which becomes 0). This leaves 7 terms: 1 with 4 derivatives of M , becoming the LM_2 term, and 6 with double derivatives of M and double derivatives of the rest. Using the above identities gives the final result.

APPENDIX B

Nucleon model

In the one-pion approximation, the nucleon-pion system is

$$\Psi \begin{pmatrix} \psi_N \\ \psi_{N\pi} \end{pmatrix}. \quad (\text{B.1})$$

It is described by the Hamiltonian

$$H = \begin{pmatrix} \hat{K}_N + m_N & W^\dagger \\ W & \hat{K}_N + \hat{K}_\pi + m_N + m_\pi \end{pmatrix}, \quad (\text{B.2})$$

Where m_N is the mass of the bare nucleon. Subtracting this mass from both sides of the Schödinger equation

$$H\Psi = E\Psi \quad (\text{B.3})$$

reduces the energy on the right hand side to the binding energy of the virtual pion. In the frame of the nucleon, its kinetic energy is 0. The Hamiltonian then reduces to

$$H = \begin{pmatrix} 0 & W^\dagger \\ W & \hat{K}_N + \hat{K}_\pi + m_\pi \end{pmatrix}. \quad (\text{B.4})$$

The coupling operator is

$$W = (\vec{\tau}\vec{\pi})(\vec{\sigma}\vec{r})F(r) \quad (\text{B.5})$$

r is the separation vector between the nucleon and the pion, and $F(r)$ is a short range form factor

$$F(r) = S_w e^{-r^2/b_w^2}, \quad (\text{B.6})$$

which is normalized by

$$S_w \rightarrow \left(4\pi \frac{3\sqrt{\pi}b_w^5}{2^{11/2}} \right) S_w. \quad (\text{B.7})$$

If the bare nucleon state is chosen as

$$\psi_N = N \uparrow, \quad (\text{B.8})$$

where N is the isospin state of the nucleon and $\uparrow$ the spin state, then the $N\pi$ -state will have the form

$$\psi_{N\pi} = (\vec{\tau}\vec{\pi})(\vec{\sigma}\vec{r})N \uparrow \sum_{i=1}^n c_i e^{-\alpha_i r^2} \quad (\text{B.9})$$

In the correlated Gaussian basis. The overlap matrix is

$$\mathcal{N} = \begin{pmatrix} \langle \psi_N | \psi_N \rangle & 0 \\ 0 & \langle \psi_i^\pi | \psi_j^\pi \rangle \end{pmatrix}, \quad (\text{B.10})$$

where ψ_i^π is the i 'th term of eq. (B.9). Then,

$$\langle \psi_N | \psi_N \rangle = 1, \quad (\text{B.11})$$

$$\langle \psi_i^\pi | \psi_j^\pi \rangle = \frac{9}{2} \frac{\pi^{3/2}}{(\alpha_i + \alpha_j)^{5/2}}. \quad (\text{B.12})$$

The Hamiltonian matrix is

$$\mathcal{H} = \begin{pmatrix} 0 & \langle \psi_N | W^\dagger | \psi_j^\pi \rangle \\ \langle \psi_i^\pi | W | \psi_N \rangle & \langle \psi_i^\pi | \hat{K}_N + \hat{K}_\pi + \hat{m}_\pi | \psi_j^\pi \rangle \end{pmatrix}. \quad (\text{B.13})$$

where

$$\langle \psi_i^\pi | W | \psi_N \rangle = \frac{9}{2} S_w \frac{\pi^{3/2}}{(\alpha_i + 1/b_w^2)^{5/2}} \quad (\text{B.14})$$

and

$$\langle \psi_i^\pi | \hat{K}_N + \hat{K}_\pi + \hat{m}_\pi | \psi_j^\pi \rangle = 45 \frac{\hbar^2}{2\mu_{N\pi}} \frac{\alpha_i \alpha_j \pi^{3/2}}{(\alpha_i + \alpha_j)^{7/2}}. \quad (\text{B.15})$$

$\mu_{N\pi} = m_N m_\pi / (m_N + m_\pi)$ is the reduced mass. Using quasi-random stochastic sampling, the energy converges very quickly with only a few Gaussians as seen on figure B.1.

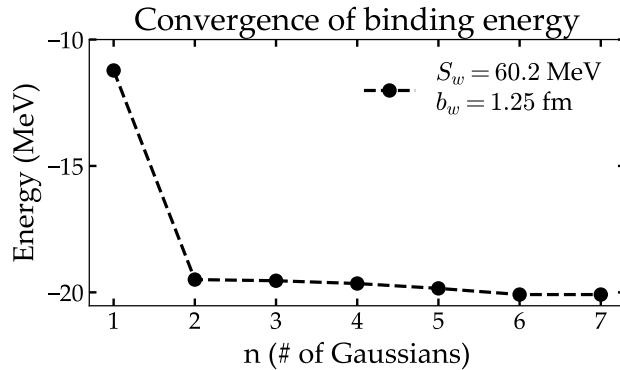


Figure B.1: Convergence of the binding energy between the nucleon and the virtual pion. The energy converges to -20.1 MeV after 6 Gaussians.